FULL SYMMETRIC TUTTE MONODROMY: HYPERCUBES, UNIFORM MATROIDS, AND DENSITY CRITERIA

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Abstract. We prove that the arithmetic and geometric Tutte monodromy groups of every hypercube Q_n, $n \geq 2$, are S_{2^n-1}, and those of every uniform matroid $U_{r,e}$, $2 \leq r < e$, are S_r. These establish the Bohn–Cameron–Müller conjecture for both families. A density criterion also gives full symmetric monodromy for connected noncomplete arc-transitive simple graphs G with $\gcd(|V(G)|-1, |E(G)|) = 1$. For odd prime-power rank, we give a primitivity criterion by excluding a finite list of candidate congruence-block trace identities. Finite-field certificates also establish full symmetric monodromy for the Coxeter and Hoffman–Singleton graphs.

The hypercube proof uses integral root expansions to compare inertia at infinity with the residual action at the dyadic Gauss place. A weighted deformation determines infinity inertia and centers block sums. Reduction modulo two realizes the roots in a vector space, where inertia and a regular unipotent Frobenius exclude the remaining block systems. A simple branch point gives a geometric transposition. For uniform matroids, binomial identities show that the discriminant is squarefree, of degree $(e-r)(r-1)$, so the finite branch transpositions generate the monodromy group. We also prove a separability and discriminant criterion for Brylawski polynomials with binomial weighted initial form. All full monodromy conclusions hold over every field of characteristic zero.

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1. Introduction

The Tutte polynomial of a graph or matroid encodes much of its combinatorial structure. Its evaluations count bases and spanning trees, and its specializations include the chromatic and flow polynomials and the random-cluster partition function [20]. These specializations reflect its universality for deletion–contraction [3]. We study its roots through their Galois group, an approach also used for chromatic polynomials [4]. For the bivariate Tutte polynomial, we take the Galois group in one variable over the rational function field in the other; this is its monodromy group. Bohn, Cameron and Müller conjectured that for every connected matroid this group is the full symmetric group. We prove the conjecture for all hypercubes and all connected uniform matroids, and develop density and trace criteria for further graph and matroid families.

Let $\mathcal{M} = (\mathcal{E}, r_{\mathcal{M}})$ be a matroid, where $r_{\mathcal{M}}$ is its rank function. Its Tutte polynomial is

$$T_{\mathcal{M}}(X,Y) = \sum_{A \subseteq \mathcal{E}} (X-1)^{r_{\mathcal{M}}(\mathcal{E}) - r_{\mathcal{M}}(A)} (Y-1)^{|A| - r_{\mathcal{M}}(A)}. \tag{1.1}$$

If $\mathcal{M}$ is loopless and coloopless, it is monic of degree $r_{\mathcal{M}}(\mathcal{E})$ in X and of degree $|\mathcal{E}| - r_{\mathcal{M}}(\mathcal{E})$ in Y. Projection of the curve $T_{\mathcal{M}}(X,Y) = 0$ to the Y-line is therefore a generically finite cover of degree $r_{\mathcal{M}}(\mathcal{E})$. Over a field K, its Galois group over the function field $K(Y)$ of the base is the arithmetic monodromy group; after extending the constants to an algebraic closure one obtains the geometric monodromy group. Over $\mathbb{C}$, the latter agrees with the topological monodromy of the cover over its unbranched locus [12, §1].

Bohn, Cameron and Müller conjectured that the monodromy group is the full symmetric group for every connected matroid of positive rank over every field of characteristic zero [2, Conjecture 9]. Their multivariate theorem [2, Theorem 1] does not imply the conjecture for the ordinary bivariate specialization. They verified the conjecture computationally for every biconnected graph on at most ten vertices; this includes Q_3 but not Q_4 [2, preceding Conjecture 9]. Goodall, Jouve and Sereni subsequently proved the conjecture for cycles [10, Theorem 4.5]. They also proved full symmetric monodromy over $\mathbb{Q}(Y)$, in each fixed rank at least two, for all sufficiently large uniform matroids [10, Theorem 4.7(1)]. Their results for a cycle with one thick edge give either a transposition or full symmetry under the additional hypotheses in [10, Theorems 4.9 and 4.11]. The present proof obtains hypercube primitivity from two places of the same bivariate cover. For uniform matroids, the critical-value calculation in Section 9 removes the size threshold and also determines geometric monodromy. It uses the Morse-polynomial approach discussed in [10, §4.2].

Write Q_n for the Cayley graph of $\mathbb{F}_2^n$ with respect to the standard basis. It has

$$M = 2^n, \qquad E = n2^{n-1}, \qquad N = M - 1, \qquad \sigma = E - N = (n-2)2^{n-1} + 1.$$

Here N and σ are the rank and nullity of its cycle matroid. Put $T_n = T_{Q_n}$, and let Ω_n be its N roots in X. We write

$$G_n^{\mathrm{ar}} = \mathrm{Gal}\big(T_n(X,Y)/\mathbb{Q}(Y)\big),$$
$$G_n^{\mathrm{geo}} = \mathrm{Gal}\big(T_n(X,Y)/\overline{\mathbb{Q}}(Y)\big).$$

In the notation of Goodall–Jouve–Sereni, these are respectively $G_{\mathbb{Q},Y}(T_n)$ and $G_{\overline{\mathbb{Q}},Y}(T_n)$.

Theorem 1.1. *For every $n \geq 2$,*

$$G_n^{\mathrm{ar}} = G_n^{\mathrm{geo}} = S_{2^n-1}. \tag{1.2}$$

The equality of the arithmetic and geometric groups says in addition that the splitting field over $\mathbb{Q}(Y)$ is regular: its constant field is $\mathbb{Q}$. Because the extension is regular, base change gives the same symmetric group over $K(Y)$ for every field K of characteristic zero (Remark 2.2). By Hilbert irreducibility [19, Proposition 3.3.5], for each fixed $n \geq 2$, the specialization $T_n(X, y_0)$ has Galois group S_{2^n-1} over $\mathbb{Q}$ for every $y_0 \in \mathbb{Q}$ outside a thin set.

Our second family result has no coprimality or size restriction.

Theorem 1.2 (uniform matroids)**.** *For every $2 \le r < e$, the arithmetic and geometric monodromy groups of $T_{U_{r,e}}(X,Y)$ are both S_r, over every field of characteristic zero. Moreover,* $\operatorname{disc}_X T_{U_{r,e}}(X,Y)$ *is squarefree of degree $(e-r)(r-1)$.*

The density argument gives a further criterion when rank and size are coprime. For a matroid of rank r, size e, and nullity $s = e - r$, write $\delta_{\mathcal{M}}(A) = e\,r_{\mathcal{M}}(A) - r|A|$ for the density defect. Writing $s_{\mathcal{M}}(A) = |A| - r_{\mathcal{M}}(A)$ for the nullity of A, the defect is the determinant of the rank–nullity profiles $(r_{\mathcal{M}}(A), s_{\mathcal{M}}(A))$ and (r,s) of A and of $\mathcal{M}$, so it is positive exactly when A has a smaller nullity-to-rank ratio than $\mathcal{M}$; see (3.3). For a connected matroid, strict uniform density means that this defect is positive for every nonempty proper subset of the ground set [6, Theorem 2.10].

Theorem 1.3 (coprime density criterion)**.** *Let $\mathcal{M}$ be a connected matroid of rank $r \ge 2$, size e, and positive nullity $s = e - r$. Suppose* $\gcd(r,e) = 1$ *and every nonempty nonspanning set A satisfies $\delta_{\mathcal{M}}(A) \ge s$, with equality exactly for singletons. Then the arithmetic and geometric monodromy groups of $T_{\mathcal{M}}(X,Y)$ are both S_r. The same conclusion holds over $K(Y)$ for every field K of characteristic zero.*

The defect hypothesis forces simplicity: a loop has defect $-r$, and a parallel pair has defect $s - r < s$. It also forces strict uniform density: nonempty nonspanning sets have positive defect, while a proper spanning set has defect $r(e - |A|) > 0$.

Corollary 1.4 (arc-transitive graphs)**.** *Let G be a connected noncomplete arc-transitive simple graph with M vertices and E edges. If* $\gcd(M-1, E) = 1$*, then the Galois group of $T_G(X,Y)$ over $K(Y)$ is S_{M-1} for every field K of characteristic zero.*

For a d-regular graph, $E = dM/2$ and coprimality is equivalently

$$\gcd(M-1, E) = \begin{cases} \gcd(M-1, d), & M \text{ even}, \\ \gcd(M-1, d/2), & M \text{ odd}. \end{cases}$$

Indeed, $M/2$ is coprime to $M-1$ when M is even; when M is odd, d is even and M is coprime to $M-1$. Thus every graph in Corollary 1.4 of even order and power-of-two valency qualifies. Corollary 8.1 covers bipartite vertex-transitive graphs of the same valencies. In valency three, the condition is $M \not\equiv 1 \pmod 3$.

Projective geometries $\mathrm{PG}(r-1, q_0)$ with $\gcd\big(r, (q_0^r - 1)/(q_0 - 1)\big) = 1$ also satisfy the criterion (Corollary 8.2); this includes the Fano matroid.

Examples include cycles C_M with $M \ge 4$, the families $K_{t,t}$ and $K_{t,t,t}$ with $t \ge 2$, and the four noncomplete Platonic graphs; further examples appear in Appendix A. For cycles it recovers the noncomplete cases of [10, Theorem 4.5] in characteristic zero by a different route; in particular, $Q_2 = C_4$ is already covered by the cycle case. The density criterion and its graph and projective-geometry consequences are proved in Section 8. For odd prime-power rank, the cyclotomic multipliers restrict every possible block system to a congruence partition. Theorem 8.4 uses its exact block traces to give a primitivity criterion for every value of $\gcd(r,e)$, including $r \mid e$. This trace criterion allows nonsingleton subsets to attain the minimum density defect. The case $\gcd(r,e) = p$ gives new proofs of S_9 for Petersen and its complement $J(5,2)$ (Corollary 8.5). Finite-field specializations give a second route to primitivity: together with a geometric transposition they yield S_{15} for Clebsch, and with Jordan's theorem and geometric discriminant certificates they yield S_{27} for Coxeter and S_{49} for Hoffman–Singleton (Corollaries 8.7 and 8.9). Petersen and its complement are already within the computational range of [2]; here they illustrate the trace argument. The graph and matroid families established in the paper are collected in Appendix A.

Alongside these monodromy results, we isolate a separability consequence of the binomial Newton face and the Brylawski identity. Following [10], a polynomial $F \in \mathbb{Z}[X,Y]$ satisfying

$$(Y-1)^r F\left(\frac{Y}{Y-1}, Y\right) = Y^{r+s} \tag{1.3}$$

is a normalized $(r+s,r)$-Brylawski polynomial. For positive weights, $\mathrm{in}_{(s,r)} F$ denotes the sum of the monomials on the corresponding maximal face of the Newton polygon.

Theorem 1.5 (Newton–Brylawski separability)**.** *Let $r, s > 0$ and let $F \in \mathbb{Z}[X,Y]$ be monic in X, satisfy (1.3), and have $\mathrm{in}_{(s,r)} F = X^r + Y^s$. Put $g = \gcd(r,s)$, and let ℓ be an odd prime such that*

$$\ell \mid r, \qquad \ell \nmid s, \qquad g < \ell. \tag{1.4}$$

Then $F \bmod \ell$ is separable in X over $\mathbb{F}_\ell(Y)$. If, moreover, r and $v_\ell(r)$ are odd, then

$$\mathrm{disc}_X F(X,Y) \notin \overline{\mathbb{Q}}(Y)^{\times 2}. \tag{1.5}$$

The separability conclusion permits good specializations and their Frobenius factor patterns. The discriminant conclusion says that geometric monodromy contains an odd permutation, because the discriminant of a separable polynomial is a square precisely when its root action lies in the alternating group. Theorem 1.5 is independent of Theorem 1.1; it applies when the first root-dependent correction is unavailable.

1.1. **Proof strategy.** The trace arguments use the density bound to control the entire polar part of the roots at infinity: the branches share a common polar series multiplied by their roots-of-unity labels. Cancellation of those labels in a block sum therefore cancels every polar term. Classical constancy and integrality arguments then force an exact block-trace identity (Theorem 7.1). The defect bound thus converts a finite phase calculation into a divisibility constraint on the number of blocks; higher coefficients can exclude the remaining candidate trace identities. For the hypercube proof, set

$$g = \gcd(N, E), \qquad L = N/g, \qquad h = \sigma/g.$$

The integer g is the number of inertia orbits above $Y = \infty$, each of length L. The proof separates into the cases $g = 1$ and $g > 1$; when N is prime, transitivity alone gives primitivity. Since N prime forces $g = 1$ (then n is prime and $N \equiv 1 \pmod{n}$), the three regimes are nested:

$$\underset{\text{transitivity}}{\{N \text{ prime}\}} \subset \underset{\text{centered sums}}{\{g = 1\}} \subset \underset{\text{characteristic two}}{\{g \geq 1\}}$$

The labels name the ingredient that primitivity needs at each level beyond the previous one; the transposition of Section 4 is common to all three. When N is prime, that is, a Mersenne prime, a transitive group of prime degree is primitive, and Theorem 1.1 follows from irreducibility, the simple branch point, and Lemma 7.7. When $g = 1$ and N is composite, the centered block sums of Section 7.1 exclude every nontrivial block system; this is the argument of Theorem 1.3, and reduction modulo two is unnecessary. When $g > 1$, which first happens at $n = 6$, centered sums give the divisibility condition $k \mid g$, and the characteristic-two action of Sections 5 and 6 forces the number of blocks to be coprime to g. Table 2.2(a) records the parameters in small dimensions; Remark 7.6 works through the three block-system tests for $n = 6$.

At infinity we shift to the rank-generating polynomial and rescale by the density defect. Strict uniform density leaves the binomial initial face $X^N + Y^\sigma$, which gives the inertia cycles. Edge isoperimetry then shows that single edges are the unique nonempty nonspanning subsets contributing to the first root-dependent correction. Its coefficient E has two uses: together with the prime supplied by Lemma 2.1, it produces a simple branch point; it also centers the sums attached to a hypothetical block system.

The second local picture, in Section 5, uses random-cluster coordinates and the 2-Gauss valuation. Nontrivial orbits of the translation group $\mathbb{F}_2^n$ on edge subsets have even cardinality, so only

translation-invariant subsets contribute modulo two. Restoring the zero root then gives a linearized polynomial. Its nonzero roots form the nonzero vectors of a space over a finite field of characteristic two. Integral expansions at infinity identify this space with the reductions of the characteristic-zero roots. In this one root frame, inertia S acts by multiplication by an element of order L, and Frobenius at $v = 1$ acts as a regular unipotent Frob. Its single Jordan block gives a one-dimensional fixed space.

For a candidate block system whose number of blocks divides g, a Sylow argument makes the induced action of $\langle S, \mathrm{Frob}\rangle$ have odd order. Thus the 2-power-order element Frob fixes every block setwise. Each block has odd size, so it contains a nonzero fixed vector; these fixed vectors form one orbit of the nonzero scalars, which belong to $\langle S\rangle$. Consequently inertia is transitive on the blocks. The affine classification then makes their number coprime to g, while centered sums make it divide g. Thus there is one block. Primitivity uses the infinity and dyadic Gauss places. A separate p-adic construction produces a simple branch point whose inertia generator is a geometric transposition. Primitivity and normality give both groups in (1.2). Figure 1.1 records the dependencies.

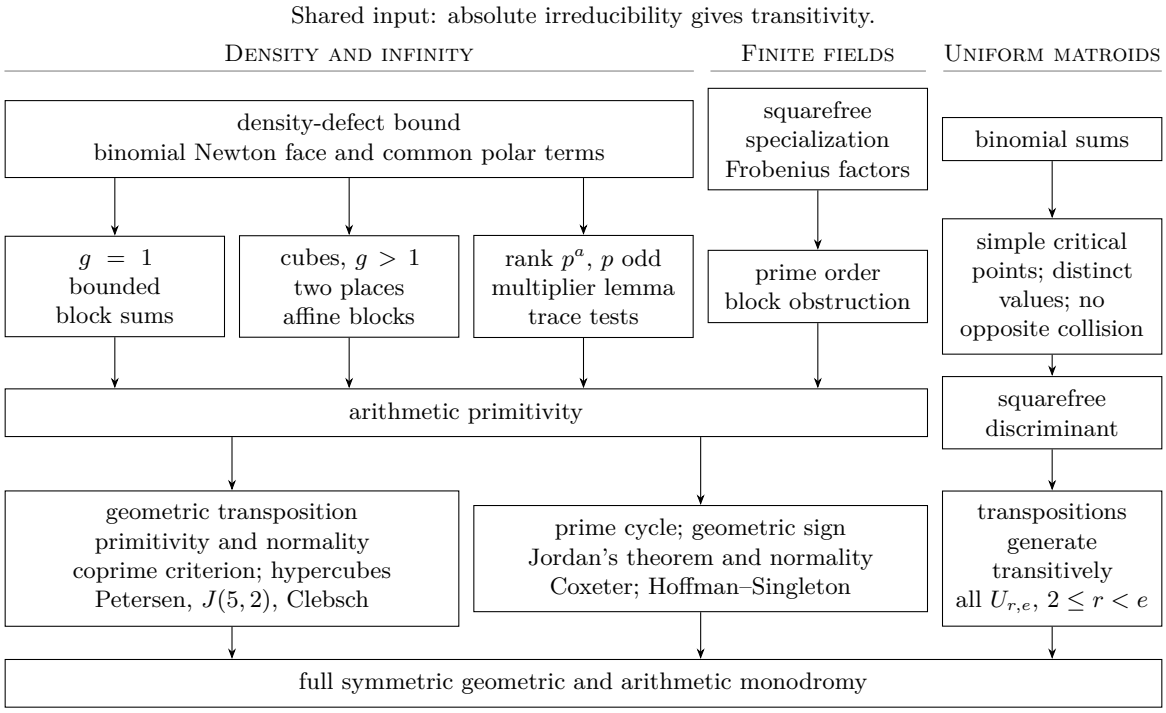


FIGURE 1.1. Proof dependencies. Here $g = \gcd(r, e)$, with $(r, e) = (N, E)$ for the cube. In its $g > 1$ route, Lemma 6.4 supplies block stabilizers before the block-trace theorem is applied. Theorem 7.1 supplies the common trace identity. The prime-power route uses Lemma 8.3 and Theorem 8.4. The final steps use additional geometric input: the simple-branch construction with singleton uniqueness (with Lemma 2.1 supplying the cube's prime), residue certificates for Petersen and $J(5, 2)$, or the normalized-discriminant certificates for Coxeter and Hoffman–Singleton. Clebsch uses the simple-branch construction with $p = 3$. The independent uniform-matroid route uses finite branch transpositions.

1.2. **Organization.** Section 2 fixes notation for the cover and its root coordinates. Section 3 develops the weighted deformation, the characteristic-zero root expansions, and the decomposition group at infinity. Section 4 constructs the independent simple branch point. Section 5 establishes integral expansions at the dyadic Gauss place, determines the characteristic-two linearized polynomial, and compares its root action with the characteristic-zero action. Section 6 studies invariant block systems for the affine group. Section 7 establishes the common block-trace theorem, then proves primitivity and completes the hypercube monodromy calculation. Section 8 proves the density criterion, the prime-power trace criterion, and the graph applications. Section 9 proves Theorem 1.2 by critical values. Its proof can be read independently of Sections 3–8, using absolute irreducibility and generation by the finite branch transpositions. Section 10 proves the independent separability and discriminant theorem. Appendix A gives a coverage table, and Appendix B records the finite certificates used for the graph applications.

For Theorem 1.3, a shorter route runs from the density expansion in Proposition 3.7 and the branch construction in Proposition 4.1 to Corollary 7.2 and the proof in Section 8. Of Sections 5–6, only the elementary Lemma 6.1 is used along this route.

Use of artificial intelligence. This work was developed by the author collaboratively with ChatGPT and Claude tools aiding mathematical exploration, computation, drafting, and editing.

2. The Tutte cover and changes of root coordinate

2.1. **Graph and matroid notation.** We identify the vertices of Q_n with $\mathbb{F}_2^n$ and join x to $x+e_i$ for $1 \le i \le n$. For any edge subset A, let $c(A)$ be the number of connected components of the spanning subgraph $(\mathbb{F}_2^n, A)$, with all vertices retained. Its graphic rank is

$$r(A) = M - c(A).$$

Its nullity is

$$s(A) = |A| - r(A).$$

For every connected matroid $\mathcal{M}$, the Tutte polynomial $T_{\mathcal{M}}$ is absolutely irreducible by the theorem of Merino, de Mier and Noy [17, Theorem 1 and the following paragraph, p. 299]; see also the UFD formulation in [10, Corollary 2.7]. Consequently its geometric, and hence its arithmetic, monodromy group is transitive on the roots. The cycle matroid of Q_n is connected when $n \ge 2$, so G_n^{geo} and G_n^{ar} are transitive on the N roots. The automorphism group of the cube is transitive on edges.

We use two transforms of T_n. The rank-generating polynomial, obtained by shifting the Tutte variables, is

$$R_n(U,v) = T_n(U+1, v+1) = \sum_{A \subseteq E(Q_n)} U^{N-r(A)} v^{s(A)}. \tag{2.1}$$

The reduced random-cluster polynomial is

$$\widehat{Z}_n(Q,v) = v^N R_n(Q/v, v) = \sum_{A \subseteq E(Q_n)} Q^{c(A)-1} v^{|A|}. \tag{2.2}$$

The usual random-cluster polynomial is $Q\widehat{Z}_n$; division by Q is exact because every spanning subgraph has at least one component. Both R_n and $\widehat{Z}_n$ are monic of degree N in their first variable. For the random-cluster/Tutte change of variables, see Sokal [20]. Over $\mathbb{Q}(v)$ their roots correspond by

$$v = Y - 1, \qquad Q = v(X-1). \tag{2.3}$$

Over $v \neq 0$, these are coordinate descriptions of the same Tutte cover and carry the same arithmetic and geometric root actions.

The Newton polygon of $F = \sum f_{ij} X^i Y^j$ is the convex hull in $\mathbb{R}^2$ of the exponent vectors (i,j) for which $f_{ij} \neq 0$. For positive weights α, β, the linear functional $(i,j) \mapsto \alpha i + \beta j$ reaches its maximum

on a face of this polygon. We write $\mathrm{in}_{(\alpha,\beta)} F$ for the sum of precisely the monomials whose exponent vectors lie on that face.

2.2. **Notation dictionary.** The symbols e, r, s denote size, rank, and nullity for an arbitrary matroid of positive rank. The corresponding cube quantities are E, N, σ. When $\mathcal{M}$ is fixed, we abbreviate $r_{\mathcal{M}}(A)$ to $r(A)$.

For the cube, write $n = am$, where $a = 2^{v_2(n)}$ and m is odd, and put $q = 2^a$. With the parameters collected in Table 2.1, we have

$$g = \underbrace{\gcd(N,\sigma)}_{\text{inertia}} = \underbrace{\gcd(N,E)}_{\text{block divisibility}} = \underbrace{\gcd(N,n)}_{\text{cube dimension}} = \underbrace{\gcd(q^m-1,m)}_{\text{residual bounds}}.$$

Indeed, $E = N + \sigma$ and $2\sigma = (n-2)N + n$, while N is odd; these identities give the first three gcd equalities. Since a is a power of two and $N = q^m - 1$ is odd, removing the factor a gives the last.

The first expression gives coprime weights $L = N/g$ and $h = \sigma/g$; Corollary 3.2 identifies g inertia cycles, each of length L. The second gives the additive order L of E modulo N. Centered block sums use this order to force the number of blocks in a putative nontrivial invariant block system to divide g (7.8). The third shows that $g = 1$ when n is a power of two. Finally, the following lemma uses $g \mid m$ to obtain a prime $p > g$ with $\mathrm{ord}_p(q) = m$ and verifies the arithmetic hypothesis of the translation-stabilizer argument.

Lemma 2.1 (arithmetic of the cube parameters)**.** *Let $n \geq 2$, with $n = am$, $q = 2^a$, $N = 2^n - 1 = q^m - 1$, $g = \gcd(N, m)$ and $L = N/g$ as above. Then:*

(i) *g is odd;*
(ii) *for every prime $\ell \mid g$, $v_\ell(N) > v_\ell(g)$; in particular every prime dividing g divides L;*
(iii) *there is an odd prime $p \mid L$ such that $\mathrm{ord}_p(q) = m$, $p > g$, and $v_p(L) = v_p(N)$;*
(iv) *$q - 1 \mid L$;*
(v) *$\mathrm{ord}_L(q) = m$.*

Proof. The first assertion follows because N is odd. For integers $x \geq 2$, $t \geq 1$ and an odd prime $\ell \mid x - 1$, we use the lifting-the-exponent identity

$$v_\ell(x^t - 1) = v_\ell(x - 1) + v_\ell(t). \tag{2.4}$$

If $\ell \mid g$, put $f_\ell = \mathrm{ord}_\ell(q)$. Then $f_\ell \mid m$ and $f_\ell \mid \ell - 1$, so $\ell \nmid f_\ell$ and

$$v_\ell(N) = v_\ell(q^{f_\ell} - 1) + v_\ell(m) > v_\ell(m) \geq v_\ell(g).$$

This proves (ii).

For $m > 1$, the Bang–Zsigmondy theorem [25] supplies a prime divisor of $q^m - 1$ that divides no $q^i - 1$ with $1 \leq i < m$. For $q \geq 2$ and $m \geq 2$, the exceptions for $q^m - 1$ are $(q, m) = (2, 6)$ and $m = 2$ with $q + 1$ a power of two; neither occurs because $m > 1$ is odd. Thus $\mathrm{ord}_p(q) = m$, so $m \mid p - 1$ and $p > m \geq g$. In particular $p \nmid g$, and the full p-part of N survives in L. If $m = 1$, then $g = 1$ and $q \geq 4$; any prime divisor of $q - 1 = L$ has all the properties in (iii).

For (iv), let $\ell_0 \mid q - 1$ be prime. Since q is even, ℓ_0 is odd, and (2.4) gives $v_{\ell_0}(q^m - 1) = v_{\ell_0}(q - 1) + v_{\ell_0}(m)$. Thus $v_{\ell_0}(g) = v_{\ell_0}(m)$ and $v_{\ell_0}(L) = v_{\ell_0}(q - 1)$, proving $q - 1 \mid L$. For (v), since $p \mid L \mid q^m - 1$, the order of q modulo L divides m and is divisible by $\mathrm{ord}_p(q) = m$. □

Table 2.2(b) lists admissible primes for $n \leq 12$. The prime p of (iii) is the one used for the cube in Sections 4 and 7.

The coordinate systems used below are summarized in Table 2.3.

2.3. **Arithmetic and geometric monodromy.** Let $\mathscr{L}_n$ be the splitting field of $T_n(X,Y)$ over $\mathbb{Q}(Y)$. The arithmetic group is $\mathrm{Gal}(\mathscr{L}_n/\mathbb{Q}(Y))$. Let $K_n^{\mathrm{const}} = \mathscr{L}_n \cap \overline{\mathbb{Q}}$ be its constant field. After choosing an algebraic closure, the geometric group identifies with $\mathrm{Gal}(\mathscr{L}_n/K_n^{\mathrm{const}}(Y))$, and there is an exact sequence

$$1 \longrightarrow G_n^{\mathrm{geo}} \longrightarrow G_n^{\mathrm{ar}} \longrightarrow \mathrm{Gal}(K_n^{\mathrm{const}}/\mathbb{Q}) \longrightarrow 1.$$

In particular

$$G_n^{\mathrm{geo}} \triangleleft G_n^{\mathrm{ar}}. \tag{2.5}$$

Remark 2.2 (base change). If $G_n^{\mathrm{geo}} = G_n^{\mathrm{ar}}$, then $K_n^{\mathrm{const}} = \mathbb{Q}$, so $\mathscr{L}_n$ is regular over $\mathbb{Q}$. A regular extension is linearly disjoint from every constant field extension, so for every field K of characteristic zero the splitting field of T_n over $K(Y)$ is $\mathscr{L}_n K(Y)$, with Galois group G_n^{ar}. The same applies to any matroid in place of Q_n.

We recall the following standard good-reduction lemma; see [18, Chapter I, §7]. We indicate the proof.

Lemma 2.3 (good reduction and specialization)**.** *Let (K,ν) be a discretely valued field with valuation ring R and residue field κ_R, and let $f \in R[X]$ be monic with $\mathrm{disc}(f) \in R^\times$. After passing to the henselization of K, choose a prolongation of ν to the splitting field of f. Reduction gives a bijection between the roots of f and those of $\bar{f}$, inertia is trivial, and reduction identifies the decomposition group with the Galois group of the residue splitting field, compatibly with these permutation actions.*

Proof. The unit discriminant makes the root algebra finite étale over R. Finite étale algebras over a henselian local ring are equivalent to finite étale algebras over its residue field [21, Lemma 10.153.7, Tag 04GK]. Applying this to a splitting algebra gives the root bijection, trivial inertia, and the stated decomposition-group identification. □

The related global specialization principle says that the Galois group of a separable specialization embeds as a permutation subgroup of the generic Galois group; see [2, Proposition 4] and [10, Proposition 4.1]. We use the stronger local form of Lemma 2.3 to compare the characteristic-two and characteristic-zero actions on the same roots; see the first application in the proof of Lemma 5.1.

Part I. Local constructions

3. Weighted deformation and ramification at infinity

3.1. **The weighted deformation in two coordinate systems.** Two changes of coordinates expose different information carried by the same Tutte cover. Rank-generating coordinates retain the rank and nullity of each subset, and a weighted rescaling makes the normalized density defect the z-adic order of its term. Random-cluster coordinates retain the number of components and the edge count, so translation symmetry can be applied coefficient by coefficient after reduction modulo two. The root coordinate change $Q = vU$ identifies the two actions wherever $v \neq 0$.

Let $\mathcal{M} = (\mathcal{E}, r_{\mathcal{M}})$ be a matroid of positive rank. With the general notation of Table 2.1, put

$$\delta_{\mathcal{M}}(A) = e\, r(A) - r|A| \tag{3.1}$$

and

$$g = \gcd(r,s), \qquad L = \frac{r}{g}, \qquad h = \frac{s}{g}, \qquad D = \frac{e}{g} = L + h. \tag{3.2}$$

The density defect is the slack in $e\, r(A) \geq r|A|$. Because g divides both e and r, its normalized value $\delta_{\mathcal{M}}(A)/g$ is an integer. With the nullity $s(A) = |A| - r(A)$ of Table 2.1, the defect is the determinant of the rank–nullity profiles of A and of $\mathcal{M}$,

$$\delta_{\mathcal{M}}(A) = s\, r(A) - r\, s(A) = \det \begin{pmatrix} r(A) & s(A) \\ r & s \end{pmatrix}, \tag{3.3}$$

TABLE 2.1. General matroid notation and its graphic specialization to the cube.

Quantity	General matroid $\mathcal{M} = (\mathcal{E}, r_{\mathcal{M}})$	Cycle matroid of Q_n
Object		
Ground set	$\mathcal{E} = E(\mathcal{M})$	Edge set $E(Q_n)$; vertex set $\mathbb{F}_2^n$ of size $M = 2^n$
Global profile		
Size	$e = \lvert\mathcal{E}\rvert$	$E = n2^{n-1}$
Rank	$r = r_{\mathcal{M}}(\mathcal{E})$	$N = M - 1 = 2^n - 1$
Nullity	$s = e - r$	$\sigma = E - N = (n-2)2^{n-1} + 1$
Subset profile		
Subset	$A \subseteq \mathcal{E}$	Edge subset $A \subseteq E(Q_n)$
Rank	$r(A) = r_{\mathcal{M}}(A)$, the maximum size of an independent subset of A	$r(A) = M - c(A)$, where $c(A)$ counts the components of $(\mathbb{F}_2^n, A)$
Nullity	$s(A) = \lvert A\rvert - r(A)$	Dimension of the cycle space of $(\mathbb{F}_2^n, A)$
Density defect	$\delta_{\mathcal{M}}(A) = \det\left(\begin{smallmatrix} r(A) & s(A) \\ r & s \end{smallmatrix}\right) = e\,r(A) - r\lvert A\rvert$	$\delta_n(A) = \sigma\,r(A) - N\,s(A) = E\,r(A) - N\lvert A\rvert$
Normalized profile		
Profile divisor	$g = \gcd(r, s) = \gcd(r, e)$	$g = \gcd(N, \sigma) = \gcd(N, E) = \gcd(N, n)$
Normalized rank	$L = r/g$ (the exponent in $v = z^{-L}$)	$L = N/g$
Normalized nullity	$h = s/g$ (the exponent in $W = z^h U$)	$h = \sigma/g$
Normalized size	$D = e/g = L + h$ (the exponent in $Q = z^{-D} W$)	$D = E/g = L + h$
Residual data (cube only)		
2-primary decomposition	—	$n = am$, where $a = 2^{v_2(n)}$ and m is odd; $q = 2^a$, $N = q^m - 1$
Residual weight	—	$\kappa = a2^{n-1} = E/m$, defined in Section 5.3
Residual polynomial and root space	—	$\mathcal{A}_n$ is defined in (5.4); its roots form V, an m-dimensional $\mathbb{F}_q$-space

TABLE 2.2. Cube parameters in small dimensions. In the second panel, p is a prime with the properties of Lemma 2.1(iii).

(a) Boundary parameters

n	N	E	σ	g	L	h	N prime?
2	3	4	1	1	3	1	yes
3	7	12	5	1	7	5	yes
4	15	32	17	1	15	17	no
5	31	80	49	1	31	49	yes
6	63	192	129	3	21	43	no
7	127	448	321	1	127	321	yes
8	255	1024	769	1	255	769	no
9	511	2304	1793	1	511	1793	no
10	1023	5120	4097	1	1023	4097	no
11	2047	11264	9217	1	2047	9217	no
12	4095	24576	20481	3	1365	6827	no

(b) Characteristic-two parameters

n	a	m	q	κ	p
2	2	1	4	4	3
3	1	3	2	4	7
4	4	1	16	32	3 or 5
5	1	5	2	16	31
6	2	3	4	64	7
7	1	7	2	64	127
8	8	1	256	1024	3, 5, 17
9	1	9	2	256	73
10	2	5	4	1024	11, 31
11	1	11	2	1024	23, 89
12	4	3	16	8192	7, 13

TABLE 2.3. Coordinate systems used in the proof.

Polynomial	Coordinates	Defined in	Role
T_n	(X, Y)	(1.1)	global cover
R_n	$U = X - 1,\ v = Y - 1$	(2.1)	density-defect expansion
$\widehat{Z}_n$	$Q = vU,\ v$	(2.2)	reduction modulo two
Φ_n	$W = z^h U,\ v = z^{-L}$	(3.6)	integral roots at infinity

so $\delta_{\mathcal{M}}(A) > 0$ says that A has a smaller nullity-to-rank ratio than $\mathcal{M}$. Being linear in the profile $(r(A), s(A))$, the defect is additive whenever the rank is, for instance over the components of a subgraph. It is also invariant under duality: the profile of $\mathcal{E} \setminus A$ in $\mathcal{M}^*$ is $(s - s(A),\, r - r(A))$ and that of $\mathcal{M}^*$ is (s, r), so $\delta_{\mathcal{M}^*}(\mathcal{E} \setminus A) = \delta_{\mathcal{M}}(A)$, matching $T_{\mathcal{M}^*}(X, Y) = T_{\mathcal{M}}(Y, X)$.

For the rank-generating polynomial $R_{\mathcal{M}}(U, v) = T_{\mathcal{M}}(1 + U, 1 + v)$, define

$$\mathscr{R}^{\delta}_{\mathcal{M}}(\tau, U, v) = \tau^{hr} R_{\mathcal{M}}(\tau^{-h} U, \tau^{-L} v). \tag{3.4}$$

Expanding by subsets gives

$$\mathscr{R}^{\delta}_{\mathcal{M}}(\tau, U, v) = \sum_{A \subseteq \mathcal{E}} \tau^{\delta_{\mathcal{M}}(A)/g} U^{r - r(A)} v^{s(A)}. \tag{3.5}$$

Thus the exponent of τ is exactly the normalized density defect. Equivalently, in the Newton polygon of $R_{\mathcal{M}}$ with weights (h, L), the monomial of A has weight $hr - \delta_{\mathcal{M}}(A)/g$, so $\delta_{\mathcal{M}}(A)/g$ is the height of its exponent point below the selected face; Figure 3.1 draws the cube case. If all defects are nonnegative, this is a polynomial rather than a Laurent polynomial.

The one-parameter model at the boundary $Y = \infty$ is

$$\Phi_{\mathcal{M}}(W, z) = \mathscr{R}^{\delta}_{\mathcal{M}}(z, W, 1) = z^{hr} R_{\mathcal{M}}(z^{-h} W, z^{-L}). \tag{3.6}$$

Here $v = z^{-L}$ approaches infinity as z approaches zero, while $U = z^{-h} W$. In random-cluster coordinates the same root is

$$Q = vU = z^{-D} W. \tag{3.7}$$

Changing the chosen Lth root z of v^{-1} by $z \mapsto \xi z$, with $\xi \in \mu_L$, fixes v and sends $W \mapsto \xi^h W$. Under strict uniform density this is the tame inertia action at infinity; Proposition 3.9 identifies its action on the root labels.

3.2. **Uniform density and the Newton polygon.** We first relate strict uniform density to a binomial weighted initial form. For a rank-r matroid $\mathcal{M}$ on e elements, its base polytope is

$$P(\mathcal{M}) = \operatorname{conv}\{\mathbf{1}_B : B \text{ is a base of } \mathcal{M}\} \subseteq \mathbb{R}^{\mathcal{E}}.$$

Here $\mathbf{1}_B$ is the incidence vector of B, and $(r/e)\mathbf{1}$ is the vector whose every coordinate is r/e. Relative interior means interior inside the affine hull of the polytope. Following Devriendt and Mulas [6, Theorem 2.10], call $\mathcal{M}$ *strictly uniformly dense* when $(r/e)\mathbf{1}$ lies in the relative interior of $P(\mathcal{M})$. Equivalently,

$$e\, r(A) \geq r|A| \tag{3.8}$$

for every $A \subseteq \mathcal{E}$, with equality only when A is a union of ground sets of connected components. Thus, for a connected matroid, strict uniform density says precisely that (3.8) is strict for every nonempty proper A.[1]

For graphs, Jakobson, MacRury, Norin and Turner [14, §2.1, Theorem 3] use the density inequality $s\, r(A) - r\, s(A) \geq 0$ to determine a supporting face. Jakobson, Langsetmo, Rivin and Turner [13, §5] propose the asymptotic balance $Q^r + v^e = 0$ in random-cluster coordinates as a heuristic. Here individual defect levels determine the polar parts and the first root-dependent correction.

The expansions at infinity follow the classical Newton–Puiseux method [5, §§3–4]: strict uniform density gives the binomial initial form, and a single ramified rescaling reduces the construction to Hensel lifting at simple roots.

[1] When $(r/e)\mathbf{1} \in P(\mathcal{M})$, the defect can also be counted. Edmonds' partition theorem, applied to r parallel copies of each element, gives e bases $B_1, \ldots, B_e$ in which every element occurs exactly r times, and then $\delta_{\mathcal{M}}(A) = \sum_{i=1}^{e} \big(r(A) - |B_i \cap A|\big)$ is the total number of elements needed to extend the independent sets $B_i \cap A$ to bases of $\mathcal{M}|A$; in particular $\delta_{\mathcal{M}}(\{a\}) = s$. For Q_n the B_i are E spanning trees decomposing the multigraph with every edge taken N times, and $\delta_n(A) = \sum_{i=1}^{E} \big(c(T_i \cap A) - c(A)\big)$ counts the extra components into which the trees cut A; the hypothesis of Theorem 1.3 says that single edges uniquely minimize this count among nonempty nonspanning edge sets.

Proposition 3.1 (weighted initial form under strict uniform density)**.** *Let $\mathcal{M}$ be a connected strictly uniformly dense matroid on e elements, of rank r and nullity $s = e - r$, with $r, s > 0$. Then*

$$\operatorname{in}_{(s,r)} T_{\mathcal{M}}(X, Y) = X^r + Y^s. \tag{3.9}$$

Consequently geometric inertia above $Y = \infty$ has cycle type $(r/g)^g$, where $g = \gcd(r, s)$; this denotes g disjoint cycles of length r/g.

Proof. Under strict uniform density, the constant term in z of (3.6) is $W^r + 1$. Equivalently the weighted initial form of the rank-generating polynomial is $U^r + v^s$; shifting the variables does not change its highest weighted part, proving (3.9). To determine the inertia permutation, use $r = gL$ and $s = gh$, with $\gcd(L, h) = 1$, and retain the coordinates $v = z^{-L}$ and $U = z^{-h}W$ of (3.6). After multiplication by $z^{rs/g}$, the rank-generating polynomial reduces at $z = 0$ to $W^r + 1$. Its roots are simple in characteristic zero, so Hensel lifting over $\overline{\mathbb{Q}}[\![z]\!]$ gives root expansions after this base change:

$$U = \eta z^{-h} + \text{higher powers of } z, \qquad \eta^r = -1.$$

The tame generator $z \mapsto \xi z$, with ξ a primitive Lth root of unity, sends η to $\eta\xi^{-h}$. Since h is coprime to L, each orbit has length L; the r root expansions therefore split into exactly g cycles. This is the asserted cycle type $(r/g)^g$. □

The binomial initial form also determines the leading term of the discriminant. Let $r, s > 0$ and let $F \in \mathbb{Q}[X, Y]$ be monic in X with $\operatorname{in}_{(s,r)} F = X^r + Y^s$. Put $\Delta_F(Y) = \operatorname{disc}_X F(X, Y)$ and set

$$\widetilde{F}(W, \tau) = \tau^{rs} F(\tau^{-s} W, \tau^{-r}).$$

Then $\widetilde{F}$ is monic in W and $\widetilde{F}(W, 0) = W^r + 1$. If the roots of $F(X, \tau^{-r})$ are α_i, those of $\widetilde{F}$ are $\tau^s \alpha_i$. Hence

$$\operatorname{disc}_W \widetilde{F}(W, \tau) = \tau^{sr(r-1)} \Delta_F(\tau^{-r}).$$

The value at $\tau = 0$ is the nonzero number $\operatorname{disc}(W^r + 1) = (-1)^{r(r-1)/2} r^r$. It follows that

$$\deg_Y \Delta_F = s(r - 1), \qquad \operatorname{lc}_Y \Delta_F = (-1)^{r(r-1)/2} r^r. \tag{3.10}$$

A matroid is *element-transitive* if its automorphism group acts transitively on its ground set. The standard averaging argument for edge-transitive graphs, recalled by Devriendt and Mulas [6, Example 3.5], applies equally to element-transitive matroids.

Corollary 3.2 (element-transitive matroids)**.** *Every connected element-transitive matroid of positive rank and nullity is strictly uniformly dense, so Proposition 3.1 applies to it. In particular, the cycle matroid of Q_n is strictly uniformly dense for every $n \geq 2$, and geometric inertia of T_n above $Y = \infty$ has cycle type L^g.*

Proof. Take the uniform probability measure on all bases. Element transitivity makes every marginal equal to r/e, while positivity on every basis places its barycenter $(r/e)\mathbf{1}$ in the relative interior of the base polytope. The criterion of Devriendt and Mulas [6, Theorem 2.10] gives strict uniform density. The cycle matroid of Q_n is element-transitive (Section 2.1), and Proposition 3.1 gives its inertia cycle type with $(r, s) = (N, \sigma)$. □

3.3. **The minimum density defect.** The centered block sums and the construction of a ramification transposition both use the first root-dependent correction to the binomial initial form. For this purpose we minimize the density defect over nonempty nonspanning sets, where *nonspanning* means $r(A) < r$. Spanning sets contribute terms independent of the root variable and are treated separately below. By (3.3),

$$\delta_{\mathcal{M}}(A) - s = \det \begin{pmatrix} r(A) - 1 & s(A) \\ r & s \end{pmatrix},$$

and for a non-loop $a \in A$ the first row is the profile of the contraction $(\mathcal{M}|A)/a$, which has rank $r(A)-1$ and nullity $s(A)$. Consequently, if $\mathcal{M}$ is simple of rank $r \geq 2$ with $s = e - r > 0$, singleton sets are the unique minimizers of the density defect among nonempty nonspanning sets precisely when

$$\frac{s(A)}{r(A)-1} < \frac{s}{r} \qquad \big(2 \leq r(A) < r\big), \tag{3.11}$$

that is, when every such contraction has a smaller nullity-to-rank ratio than $\mathcal{M}$. For a connected graphic restriction H with v_H vertices and e_H edges, the left side is $(e_H - v_H + 1)/(v_H - 2)$.

Figure 3.1 shows the selected face and the parallel line carrying the singleton term.

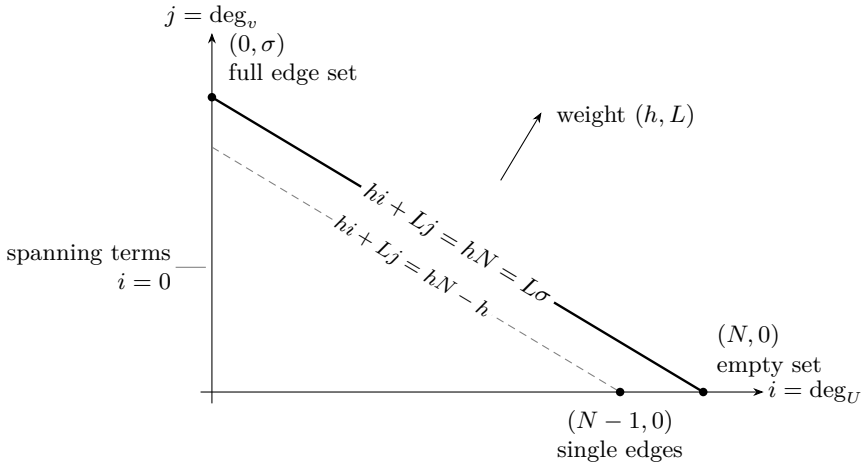


FIGURE 3.1. The selected Newton face of $R_n(U, v)$, drawn schematically. Its endpoints give $W^N + 1$ after rescaling. The dashed parallel has normalized density defect h; single edges give its only nonspanning term. Root-independent spanning terms lie on $i = 0$.

The graph-theoretic input can be expressed by the 2-isoperimetric edge-connectivity. For a graph G on at least four vertices, let ∂S be the edge cut consisting of the edges with exactly one endpoint in S, and put

$$\lambda_2(G) = \min\{|\partial S| : 2 \leq |S| \leq |V(G)| - 2\}. \tag{3.12}$$

Proposition 3.3 (density-defect bound from edge-isoperimetry)**.** *Let G be a connected noncomplete d-regular simple graph on M vertices. Using the same capital letters as for the cube, put $E = dM/2$, $N = M - 1$, and $\sigma = E - N$. If $\lambda_2(G) = 2d - 2$, then every nonempty edge set A with $r(A) < N$ satisfies*

$$\delta(A) = E\, r(A) - N|A| \geq \sigma, \tag{3.13}$$

with equality if and only if A is a single edge.

Proof. Here $d \geq 2$ and $\sigma = (d-2)M/2 + 1 > 0$, so $\sigma/N > (d-2)/2$. Decompose the subgraph with edge set A into its components containing an edge. For one such component H, write $v_H = |V(H)|$ and $e_H = |E(H)|$; its rank is $v_H - 1$ and its nullity is $e_H - v_H + 1$. If $v_H = M$, then A is spanning, contrary to $r(A) < N$. If $3 \leq v_H \leq M - 2$, then $|\partial V(H)| \geq 2d - 2$ gives

$$e_H \leq \frac{dv_H - |\partial V(H)|}{2} \leq 1 + \frac{d}{2}(v_H - 2), \qquad \text{hence} \qquad \frac{e_H - v_H + 1}{v_H - 2} \leq \frac{d-2}{2} < \frac{\sigma}{N},$$

which is (3.11) for H. If $v_H = M - 1$, then $e_H \leq E - d$, so the left side of (3.11) for H is at most $(\sigma - d + 1)/(M - 3)$, and $(\sigma - d + 1)(M - 1) < \sigma(M - 3)$ is equivalent to $M > d + 1$, which holds because G is noncomplete. For $v_H = 2$, connectedness and simplicity give one edge and equality.

The density defect is additive over the components containing an edge, while $\sigma > 0$; hence more than one component makes (3.13) strict. □

We recall the triangle-free case of Xu's restricted edge-connectivity theorem, as presented by Xu–Xu [24, Theorem 6, p. 296], in the notation of (3.12).

Lemma 3.4 (triangle-free vertex-transitive graphs)**.** *Let G be a connected triangle-free vertex-transitive simple graph of valency $d \geq 2$. Then $\lambda_2(G) = 2d-2$, and the conclusion of Proposition 3.3 holds for G.*

Proof. For $d = 2$, the graph is a cycle of length at least four and $\lambda_2(G) = 2$. For $d \geq 3$, Xu's theorem gives restricted edge-connectivity $2d - 2$, where the minimum is taken over edge cuts leaving no isolated vertices. The ordinary edge-connectivity is d: a smaller minimum cut would have two components of order at least two, contradicting the restricted bound. Consequently, a disconnected shore has boundary at least $2d$. An adjacent pair has boundary $2d-2$, so a minimizer in (3.12) has both shores connected, and the two parameters agree. Proposition 3.3 applies. □

Proposition 3.5 (minimum density defect for hypercubes)**.** *One has $\lambda_2(Q_n) = 2n - 2$. Consequently, by Proposition 3.3 with $d = n$, every nonempty edge set $A \subseteq E(Q_n)$ with $r(A) < N$ satisfies $\delta_n(A) \geq \sigma$, with equality if and only if A consists of one edge.*

Proof. The hypercube Q_n is triangle-free and vertex-transitive of valency $n \geq 2$, so Lemma 3.4 applies. □

A graph is *arc-transitive* if its automorphism group is transitive on ordered pairs of adjacent vertices.

Corollary 3.6 (density-defect bound for arc-transitive graphs)**.** *Let G be a connected noncomplete arc-transitive simple graph of valency d. Then $\lambda_2(G) = 2d-2$, and the conclusion of Proposition 3.3 holds for G.*

Proof. Since G is connected and noncomplete, $d \geq 2$. Hamidoune, Lladó, Serra and Tindell [11, Corollary 5] prove the optimal k-isoperimetric bound for arc-transitive graphs of degree $d \geq 3k - 3$. Their parameter minimizes the edge boundary over $k \leq |S| \leq |V(G)|/2$; by complement symmetry, its $k = 2$ instance is precisely $\lambda_2(G) = 2d - 2$ for $d \geq 3$, with λ_2 as in (3.12). The degree-two case consists of the cycles C_M, $M \geq 4$, for which $\lambda_2 = 2$. Proposition 3.3 applies. □

The preceding graph-theoretic input has the following polynomial consequence.

Proposition 3.7 (first root-dependent correction)**.** *Let $\mathcal{M}$ be a connected matroid of rank $r \geq 2$, size e, and positive nullity $s = e-r$. Suppose every nonempty nonspanning set A satisfies $\delta_{\mathcal{M}}(A) \geq s$. Then $\mathcal{M}$ is simple and strictly uniformly dense. Put*

$$P_{\mathcal{M}}(W) = \sum_{\substack{\varnothing \neq A \subseteq \mathcal{E},\ r(A)<r \\ \delta_{\mathcal{M}}(A)=s}} W^{r-r(A)}. \tag{3.14}$$

For the parameters in (3.2), one has

$$\Phi_{\mathcal{M}}(W,z) = W^r + C_{\mathcal{M}}^{\mathrm{span}}(z^L) + z^h P_{\mathcal{M}}(W) + z^{h+1}\mathcal{B}_{\mathcal{M}}(W,z), \qquad C_{\mathcal{M}}^{\mathrm{span}}(0) = 1. \tag{3.15}$$

If equality occurs exactly for singletons, then $P_{\mathcal{M}}(W) = eW^{r-1}$.

Proof. A loop or a parallel pair would have defect less than s, so the matroid is simple. The assumed bound gives positive defect for nonspanning sets; a proper spanning set has defect $r(e - |A|) > 0$. Thus $\mathcal{M}$ is strictly uniformly dense. In (3.5), the empty set contributes W^r. The spanning sets give precisely the terms of W-degree zero; their z-exponents are $L(e - |A|)$, and the full ground set contributes 1. The nonempty nonspanning sets of defect s give $z^h P_{\mathcal{M}}$; all other such sets have normalized defect at least $h + 1$. Each singleton contributes W^{r-1} to $P_{\mathcal{M}}$. □

Corollary 3.8 (branch expansions)**.** *Let $\mathcal{M}$ satisfy the hypotheses of Proposition 3.7. Choose $\gamma(z^L) \in \mathbb{Q}[\![z^L]\!]$ with $\gamma(0) = 1$ and $\gamma(z^L)^r = C_{\mathcal{M}}^{\mathrm{span}}(z^L)$. For every $\eta^r = -1$ there is a unique branch $W_\eta \in \mathbb{Q}(\eta)[\![z]\!]$ with $W_\eta(0) = \eta$, and*

$$W_\eta(z) = \eta\gamma(z^L) - \frac{P_{\mathcal{M}}(\eta)}{r\eta^{r-1}} z^h + O(z^{h+1}). \tag{3.16}$$

For $U_\eta = z^{-h} W_\eta$, the polar part is the negative-power part of $\eta\gamma(z^L)z^{-h}$. If $P_{\mathcal{M}}(W) = eW^{r-1}$,

$$rU_\eta + e = r\eta\gamma(z^L)z^{-h} + O(z). \tag{3.17}$$

For $\xi \in \mu_L$, substitution $z \mapsto \xi z$ fixes $v = z^{-L}$ and sends U_η to $U_{\xi^{-h}\eta}$.

Proof. The initial polynomial $W^r + 1$ has simple roots, so Hensel lifting over $\mathbb{Q}(\eta)[\![z]\!]$ gives the unique branch. The residual value at $\eta\gamma(z^L)$ has order at least h and the derivative is a unit. Hence $\theta = W_\eta - \eta\gamma$ has order at least h. Substitution in (3.15), using $\gamma(0) = 1$ and $h \geq 1$, gives

$$r\eta^{r-1}\theta = -P_{\mathcal{M}}(\eta)z^h + O(z^{h+1}).$$

This proves both expansions. By (3.6), $\Phi_{\mathcal{M}}(\xi^h W, \xi z) = \xi^{hr}\Phi_{\mathcal{M}}(W, z)$. Thus $\xi^{-h}W_\eta(\xi z)$ is the branch with constant term $\xi^{-h}\eta$, proving the asserted action by uniqueness. □

For the cube, Proposition 3.5 verifies the hypothesis of Proposition 3.7, so (3.15) holds with $(r, e, s) = (N, E, \sigma)$ and $P_{\mathcal{M}}(W) = EW^{N-1}$; we write C_n^{span} and $\mathcal{B}_n$ for the cube's series. Since Q_n has no bridges, deleting any one of its E edges leaves a spanning set, and therefore

$$C_n^{\mathrm{span}}(z^L) = 1 + Ez^L + O(z^{2L}). \tag{3.18}$$

For $n \geq 4$, one has

$$h - L = \frac{\sigma - N}{g} = \frac{(n-4)2^{n-1} + 2}{g} > 0.$$

Thus the root-independent spanning series begins before the singleton term in these dimensions. "First correction" in (3.15) always means first among the terms that depend on W.

3.4. **The decomposition group at infinity.** Corollary 3.8 applies to the cube. Since N is odd, the roots of $W^N + 1$ are $-\zeta_N^j$, $j \in \mathbb{Z}/N$; write W_j for the branch with $W_j(0) = -\zeta_N^j$. With $\gamma(z^L) \in \mathbb{Q}[\![z^L]\!]$, $\gamma(0) = 1$, $\gamma(z^L)^N = C_n^{\mathrm{span}}(z^L)$, the expansion (3.16) reads

$$W_j(z) = -\zeta_N^j\gamma(z^L) - \frac{E}{N}z^h + O(z^{h+1}), \qquad j \in \mathbb{Z}/N, \tag{3.19}$$

and the corresponding roots of R_n are

$$U_j(z) = z^{-h}W_j(z), \qquad v = z^{-L}. \tag{3.20}$$

Example ($n = 2$). For $Q_2 = C_4$, one has $N = 3$, $E = 4$, $\sigma = g = h = 1$, and $L = 3$. Thus

$$R_2(U, v) = U^3 + 4U^2 + 6U + 4 + v,$$
$$\Phi_2(W, z) = W^3 + (1 + 4z^3) + 4zW^2 + 6z^2W.$$

The full edge set and the four three-edge spanning sets give $C_2^{\mathrm{span}}(z^3) = 1 + 4z^3$; the singleton term is $4zW^2$, and the six two-edge sets give $\mathcal{B}_2 = 6W$. With $\gamma(z^3) = (1 + 4z^3)^{1/3}$ chosen to have constant term one, (3.19) becomes

$$W_j(z) = -\zeta_3^j\gamma(z^3) - \frac{4}{3}z + O(z^2), \qquad U_j(z) = z^{-1}W_j(z), \quad j \in \mathbb{Z}/3.$$

Proposition 3.9 (cyclotomic labeling of the roots)**.** *Let $\mathcal{M}$ be a connected strictly uniformly dense matroid of odd rank $r > 1$ and positive nullity $s = e - r$. Put $g = \gcd(r, e)$, $L = r/g$ and $h = s/g$, and label the branches by $W_j(0) = -\zeta_r^j$, $j \in \mathbb{Z}/r$. The splitting field of $R_{\mathcal{M}}(U, v)$ over $\mathbb{Q}((v^{-1}))$ is*

$$\mathbb{Q}(\zeta_r)((z)), \qquad z^L = v^{-1}. \tag{3.21}$$

Its arithmetic decomposition group acts on the labels by

$$\mathcal{D}_{r,g} = \{j \longmapsto uj + gb : u \in (\mathbb{Z}/r)^\times,\ b \in \mathbb{Z}/L\}. \tag{3.22}$$

The geometric inertia subgroup is the translation group $g\mathbb{Z}/r \cong C_L$.

Proof. The normalized polynomial has initial form $W^r + 1$ by Proposition 3.1. Its Hensel coefficients lie in $\mathbb{Q}(\zeta_r)$, so all the roots $U_j = z^{-h} W_j$ lie in $\mathbb{Q}(\zeta_r)((z))$. Each has valuation $-h/L$ over $\mathbb{Q}((v^{-1}))$; since $\gcd(h, L) = 1$, the local splitting field has ramification index at least L. The ratios U_j/U_0 have residues ζ_r^j, so its residue field contains $\mathbb{Q}(\zeta_r)$ and has degree at least $\varphi(r)$. The containing extension has degree $L\varphi(r)$, and the fundamental inequality forces equality.

The rotations $z \mapsto \zeta_L^b z$ and the constant automorphisms $\zeta_r \mapsto \zeta_r^u$, $z \mapsto z$, act on leading terms by translations by multiples of g and multiplication by u. Since $\gcd(h, L) = 1$, every translation in (3.22) occurs. Uniqueness of Hensel lifting makes these exact actions on the complete branches; in particular, $M_u(W_j) = W_{uj}$ and every multiplier fixes the branch labeled 0. The decomposition group embeds into arithmetic monodromy at the chosen place, so $\mathcal{D}_{r,g}$ preserves every arithmetic block system. □

For the cube, this gives $\mathcal{D}_{N,g}$ on the labels in (3.19). When $g = 1$, this group is $\mathcal{D}_{N,1} = \mathrm{AGL}_1(\mathbb{Z}/N)$.

Fix the primitive $\xi \in \mu_L$ with $\xi^{-h} = \zeta_N^g$, and write $S \in G_n^{\mathrm{ar}}$ for the element induced by $z \mapsto \xi z$ at the chosen place. It generates geometric inertia at infinity, has order L, and by Corollary 3.8 acts on the labels of (3.19) as the translation

$$T\colon j \longmapsto j + g. \tag{3.23}$$

4. A simple branch point

The following construction uses the singleton-only correction $P_{\mathcal{M}}(W) = eW^{r-1}$.

Proposition 4.1 (simple ramification)**.** *Let $\mathcal{M}$ satisfy the hypotheses of Proposition 3.7, with $\delta_{\mathcal{M}}(A) = s$ exactly for singletons. If p is any prime such that $p \mid r$ and $p \nmid g$, then there is an algebraic value $v_{\mathrm{br}} \neq 0$ for which $R_{\mathcal{M}}(U, v_{\mathrm{br}})$ has exactly one multiple root. That root is double, and local geometric inertia interchanges its two branches.*

Proof. Put $\epsilon = v_p(r)$, $r_0 = r/p^\epsilon$, and $\Lambda_1 = e(r-1)$. Choose π with $\pi^h = p$ and normalize $v_\pi(\pi) = 1$. Let $\mathcal{O}_\pi$ be the strict henselization of the valuation ring of $\mathbb{Q}_p(\pi)$, whose residue field is $\overline{\mathbb{F}}_p$. Define

$$\begin{aligned} \mathcal{F}(u, v) &= \mathscr{R}^{\delta}_{\mathcal{M}}(\pi^\epsilon, u, v) \\ &= \pi^{\epsilon h r} R_{\mathcal{M}}\left(\pi^{-\epsilon h} u, \pi^{-\epsilon L} v\right), \\ \mathcal{G}(u, v) &= \pi^{-\epsilon h} \mathcal{F}_u(u, v). \end{aligned} \tag{4.1}$$

Because $p \mid r$, the derivative of the binomial reduction $u^r + v^s$ vanishes in the root variable. Dividing by $p^\epsilon = \pi^{\epsilon h}$ retains the empty-set and singleton derivative terms. The empty-set coefficient is $r/p^\epsilon = r_0$. For a nonspanning set A with $|A| \geq 2$, the corresponding term of $\mathcal{G}$ contains the factor $\pi^{\epsilon(\delta_{\mathcal{M}}(A)/g - h)}$, whose exponent is positive by the singleton-uniqueness hypothesis. Consequently

$$\overline{\mathcal{F}}(u, v) = u^r + v^s, \qquad \overline{\mathcal{G}}(u, v) = u^{r-2}(r_0 u + \Lambda_1). \tag{4.2}$$

All coefficients displayed here are units. Indeed, $g = \gcd(r, e) = \gcd(r, s)$, so $p \nmid es$, while $p \nmid r_0(r-1)$. The nonzero common zeros of the two reduced equations satisfy

$$\bar{u}_c = -\frac{\Lambda_1}{r_0}, \qquad \bar{v}^s = -\bar{u}_c^r.$$

Choose one such $\bar{v}$. The Jacobian determinant of $(\overline{\mathcal{F}}, \overline{\mathcal{G}})$ there is

$$-sr_0\bar{u}_c^{r-2}\bar{v}^{s-1} \neq 0.$$

Multivariable Hensel lifting gives a unique common zero (u_*, v_*) above $(\bar{u}_c, \bar{v})$. Undoing the dilation gives $v_{\mathrm{br}} = \pi^{-\epsilon L} v_* \neq 0$. Since $R_{\mathcal{M}}(U, v_{\mathrm{br}})$ has a multiple root, v_{br} is a zero of $\mathrm{disc}_U R_{\mathcal{M}} \in \mathbb{Z}[v]$. This discriminant is nonzero by the separable expansions of Proposition 3.1, so v_{br} is algebraic over $\mathbb{Q}$.

At this point $\mathcal{F}_v$ and $\mathcal{G}_u$ are units. Since $\mathcal{F}_u = \pi^{\epsilon h}\mathcal{G}$, one has $\mathcal{F}_{uu} = \pi^{\epsilon h}\mathcal{G}_u \neq 0$; the critical root is exactly double. The polynomial $\mathcal{F}(u, v_*)$ is monic and integral, and its constant coefficient is a unit. Hence all its roots are units. Every repeated root reduces to a nonzero common zero of $\overline{\mathcal{F}}$ and $\overline{\mathcal{G}}$, and above the fixed $\bar{v}$ there is only $\bar{u}_c$. For fixed $v = v_*$, $\bar{u}_c$ is a simple root of $\overline{\mathcal{G}}(u, \bar{v})$, so $\mathcal{G}(u, v_*)$ has a unique root reducing to it, also in any algebraic valued extension. Hence no second collision occurs in this fiber.

Put $K = \mathrm{Frac}(\mathcal{O}_\pi)$ and $t = v - v_*$. Hensel factorization over $K[\![t]\!]$ separates the double root from the other roots, giving an identity in $K[\![t]\!][x]$:

$$\mathcal{F}(u_* + x, v_* + t) = \big(x^2 + a(t)x + b(t)\big)H_0(x, t),$$

where $a(0) = b(0) = 0$ and $H_0(0, 0) \neq 0$. Since $\mathcal{F}_v \neq 0$, one has $b'(0)H_0(0, 0) = \mathcal{F}_v(u_*, v_*) \neq 0$. The discriminant of the quadratic factor therefore has a simple zero at $t = 0$. The discriminant of the other factor and their resultant are units, so $\mathrm{disc}_U R_{\mathcal{M}}$ has a simple zero at v_{br}. Local inertia interchanges precisely these two branches. The branch value is algebraic over the constant field, placing this transposition in geometric monodromy. □

For Q_n, any prime dividing N but not g qualifies. Using the prime from Lemma 2.1(iii) in this construction gives the geometric transposition used in the proof of Theorem 1.1. Since $v_{\mathrm{br}} = \pi^{-\epsilon L} v_*$ with v_* a unit, this finite branch value reduces to infinity at the chosen p-adic place. This prime is separate from the dyadic place used for the root comparison.

5. Reduction modulo two and the residual linear action

We first show that the characteristic-zero branches at infinity are integral and identify their reduction at the dyadic Gauss place. Translation symmetry then makes the reduced random-cluster polynomial linearized after its zero root is restored. Frobenius at $v = 1$ acts as a regular unipotent element, and the integral comparison transports its action to the characteristic-zero roots. Table 5.1 records the roles of the local constructions.

5.1. **Integral expansions and the dyadic Gauss place.** At a prime $p \nmid N$, the binomial weighted initial form gives integral root expansions with two local descriptions: the z-adic one at infinity and the localization of the coefficient power-series ring at (p), where z is a unit. The latter induces the p-Gauss valuation on $\mathbb{Q}(v)$, with $v = z^{-L}$ remaining transcendental in the residue field. The construction below compares inertia at infinity with the residual Galois action through these same series.

Lemma 5.1 (integral root expansions at infinity)**.** *Let $f(U,v) \in \mathbb{Z}[v][U]$ be monic of degree N. Suppose $L \mid N$, $h > 0$, $\gcd(h,L) = 1$, and*

$$\Phi_f(W,z) = z^{hN} f(z^{-h}W, z^{-L}) \in \mathbb{Z}[W,z], \qquad \Phi_f(W,0) = W^N + 1.$$

Let $p \nmid N$, and let $\mathcal{O}_p$ be the ring of integers of a finite unramified extension of $\mathbb{Q}_p$ containing all roots $\omega^N = -1$ and μ_L. There are unique series

$$W_\omega(z) \in \mathcal{O}_p[\![z]\!], \qquad W_\omega(0) = \omega, \qquad \Phi_f(W_\omega(z), z) = 0.$$

For $v = z^{-L}$, the roots $U_\omega = z^{-h}W_\omega(z)$ and all their pairwise differences are units for the valuation defined by $\mathcal{O}_p[\![z]\!]_{(p)}$. Its restriction to $\mathbb{Q}(v)$ is the p-Gauss valuation, defined on a nonzero polynomial by $v_p^{\mathrm{G}}(\sum_i a_i v^i) = \min_i v_p(a_i)$ and extended to rational functions by taking differences of valuations. The substitutions $z \mapsto \xi z$, $\xi \in \mu_L$, preserve the restricted prolongation to the global splitting field and act on the roots by

$$U_\omega \longmapsto U_{\xi^{-h}\omega}.$$

Reduction identifies the decomposition group at this extension of the p-Gauss valuation with the Galois group of the splitting field of $\bar{f}$ over $\mathbb{F}_p(v)$, with the same permutation action on the roots.

TABLE 5.1. Local data used in the hypercube proof. The value $v = 1$ belongs to the characteristic-two reduction; v_{br} is a characteristic-zero branch value constructed p-adically.

Place	Coordinates	Structure and use	Reference
$Y = \infty$	$v = z^{-L}$, $W = z^h U$	Inertia S has g cycles of length L; the affine decomposition group $\mathcal{D}_{N,g}$ acts on the labels used for centered sums.	Prop. 3.9; Cor. 7.2
Dyadic Gauss place	$Q = z^{-D}W$, followed by reduction modulo 2	The residual roots form $V^\times$ with $V \cong \mathbb{F}_q^m$; the same inertia acts by multiplication. This identifies the two root actions.	§5.1; Prop. 5.6
$v = 1$ in characteristic 2	$\mathcal{A}_n(Q,1)$	Lifted Frobenius Frob $\sim J_m(1)$ has a one-dimensional fixed space, used to prove inertia transitive on blocks.	Prop. 5.5; Cor. 7.4
$v = v_{\mathrm{br}}$; $p \mid N$, $p \nmid g$	$u = \pi^{\epsilon h}U$, $v_{\mathrm{br}} = \pi^{-\epsilon L}v_*$	Local inertia at the simple branch is generated by a geometric transposition t_p; primitivity then gives full symmetry.	Prop. 4.1; §7

Proof. Since $p \nmid N$, the roots of $W^N + 1$ are distinct modulo p. Their derivatives $N\omega^{N-1}$ are units of $\mathcal{O}_p$. Formal Hensel lifting in the z-adically complete ring $\mathcal{O}_p[\![z]\!]$ gives the stated series and the factorization

$$\Phi_f(W,z) = \prod_{\omega^N=-1} (W - W_\omega(z)).$$

The constant terms of W_ω and of $W_\omega - W_{\omega'}$, for $\omega \neq \omega'$, are units. In the height-one localization at (p), the parameter z is also a unit. Hence the U-roots and their differences are units there.

The residue field of this localization is $\kappa_p((z))$, where $\kappa_p = \mathcal{O}_p/(p)$. A polynomial in v, after division by p to the power of the minimum p-valuation of its coefficients, reduces under $v = z^{-L}$ to a nonzero Laurent polynomial: distinct powers of v remain distinct powers of z. This proves the Gauss assertion, first for polynomials and then for rational functions.

The substitution $z \mapsto \xi z$ fixes v and $\mathcal{O}_p$, and preserves (p). From the defining weighted rescaling and $L \mid N$, $\xi^{-h} W_\omega(\xi z)$ is a root of $\Phi_f(W, z)$ with constant term $\xi^{-h}\omega$. Hensel uniqueness gives

$$W_{\xi^{-h}\omega}(z) = \xi^{-h} W_\omega(\xi z). \tag{5.1}$$

It follows that this substitution permutes the U-roots, restricts to their global splitting field, and preserves the chosen valuation. Its action has order L when ξ is primitive.

The unit root differences make the discriminant a unit in the Gauss valuation ring. Lemma 2.3, applied after henselization, identifies the decomposition group with the residue Galois group and shows that inertia is trivial. □

For the cube we take $p = 2$. Choose the place at infinity determined by $v = z^{-L}$ and the corresponding embedding ι_∞ of $\mathscr{L}_n$; after choosing a dyadic embedding of $\mathbb{Q}(\zeta_N)$, put $K_2 = \mathbb{Q}_2(\zeta_N)$ and $\mathcal{O} = \mathbb{Z}_2[\zeta_N]$. The integral branches of the lemma agree with (3.19) by Hensel uniqueness and give, inside $K_2((z))$,

$$\iota_\infty(\mathscr{L}_n) \subset \mathbb{Q}(\zeta_N)((z)) \cap \mathrm{Frac}(\mathcal{O}[\![z]\!]) \subset K_2((z)). \tag{5.2}$$

The height-one primes (z) and (2) of $\mathcal{O}[\![z]\!]$ restrict to $Y = \infty$ and to the dyadic Gauss valuation respectively, and the substitution $z \mapsto \xi z$ preserves both.

5.2. **Translation fixed points in random-cluster coordinates.**

Proposition 5.2 (parity of the reduced random-cluster polynomial)**.** *In $\mathbb{F}_2[Q, v]$ one has*

$$\overline{\widehat{Z}_n}(Q, v) = \sum_{d=0}^{n} \binom{n}{d} Q^{2^{n-d}-1} v^{d2^{n-1}}. \tag{5.3}$$

Proof. For fixed values of $c(A)$ and $|A|$, the corresponding coefficient in (2.2) counts a finite set on which $\mathbb{F}_2^n$ acts by translation. Modulo two its cardinality equals the number of points fixed by the whole translation group. A fixed edge set is a union of complete direction classes. If d directions are chosen, the resulting graph is a disjoint union of 2^{n-d} copies of Q_d and has $d2^{n-1}$ edges. There are $\binom{n}{d}$ choices. □

5.3. **The residual linearized polynomial.** Recall from Section 2.2 that $n = am$ with m odd and $q = 2^a$, so that $q^m = M$ and $N = q^m - 1$. Put $\kappa = a2^{n-1}$, a power of two with $E = m\kappa$.

A polynomial of the form $\sum_i a_i Q^{q^i}$ is called q-linearized; it defines an $\mathbb{F}_q$-linear map on every field containing $\mathbb{F}_q$ and its coefficients [16, §3.4].

Proposition 5.3 (linearized reduction)**.** *In $\mathbb{F}_2[Q, v]$, one has*

$$\mathcal{A}_n(Q, v) := Q\overline{\widehat{Z}_n}(Q, v) = \sum_{j=0}^{m} \binom{m}{j} v^{\kappa(m-j)} Q^{q^j}. \tag{5.4}$$

Over $\mathbb{F}_q(v)$, this is a separable q-linearized polynomial. Its roots form an m-dimensional $\mathbb{F}_q$-vector space V, and the roots of $\overline{\widehat{Z}_n}$ are $V^\times = V \setminus \{0\}$.

Proof. The binary digits of $n = am$ are those of m shifted by $v_2(n)$. Lucas's theorem therefore says that $\binom{n}{d}$ is odd exactly when $d = ai$ and $\binom{m}{i}$ is odd. Multiplying (5.3) by Q and replacing i by $m - j$ gives (5.4).

Every exponent of Q is a power of q, so the polynomial is additive and commutes with multiplication by $\mathbb{F}_q$. Its derivative is

$$\partial_Q \mathcal{A}_n(Q, v) = v^{m\kappa} = v^E \neq 0. \tag{5.5}$$

It has q^m distinct roots, closed under addition and $\mathbb{F}_q$-scalars, and therefore an m-dimensional root space. Division by Q removes precisely the zero root. □

The same calculation determines the reduction of the discriminant.

Lemma 5.4 (the dyadic discriminant)**.** *The polynomial* $\Delta_n(v) = \operatorname{disc}_U R_n(U, v)$ *has reduction*

$$\overline{\Delta_n}(v) = v^{(E-N)(N-1)} = v^{\sigma(N-1)}. \tag{5.6}$$

Proof. Put $f(Q, v) = \overline{\widehat{Z}_n}(Q, v)$. Since $\mathcal{A}_n = Qf$, $f(0, v) = v^E$, and $\partial_Q \mathcal{A}_n = v^E$, the product formula for discriminants gives

$$\operatorname{disc}_Q \mathcal{A}_n = v^{E(N+1)}, \qquad \operatorname{disc}_Q f = \frac{\operatorname{disc}_Q \mathcal{A}_n}{f(0, v)^2} = v^{E(N-1)}.$$

The identity $\widehat{Z}_n(Q, v) = v^N R_n(Q/v, v)$ multiplies every root difference by v. Thus (5.6) follows. □

Every specialization $v = c \in \overline{\mathbb{F}}_2^{\times}$ is therefore separable; Section 5.4 uses $v = 1$.

For $n = 3$, all four binomial coefficients in (5.4) are odd, so

$$\overline{\widehat{Z}_3}(Q, v) = Q^7 + v^4Q^3 + v^8Q + v^{12},$$
$$\mathcal{A}_3(Q, v) = Q^8 + v^4Q^4 + v^8Q^2 + v^{12}Q.$$

For $n = 6$, the natural field of linearity is $\mathbb{F}_4$: here $m = 3$, $q = 4$, and $\kappa = 64$, giving

$$\mathcal{A}_6(Q, v) = Q^{64} + v^{64}Q^{16} + v^{128}Q^4 + v^{192}Q.$$

Thus the 2-part of n is absorbed into $q = 2^a$.

5.4. **Frobenius at $v = 1$.** At $v = 1$, the polynomial $\mathcal{A}_n$ is an iterated composition of the sum of Frobenius and the identity.

Proposition 5.5 (a regular unipotent element from $v = 1$)**.** *For every odd $m \geq 1$, the Galois group of $\mathcal{A}_n$ over $\mathbb{F}_q(v)$ contains an element* Frob *acting on V with minimal and characteristic polynomial $(t + 1)^m$. Equivalently,* Frob *is regular unipotent: it has the single Jordan block $J_m(1)$.*

Proof. Let $\mathrm{Fr}_q : x \mapsto x^q$ be arithmetic q-Frobenius and put $\mathcal{N} = \mathrm{Fr}_q + \mathrm{id}$. Write $\mathcal{N}^{\circ j}$ for its j-fold composition. At $v = 1$, the binomial expansion gives the identity of additive polynomials

$$\mathcal{A}_n(Q, 1) = \mathcal{N}^{\circ m}(Q). \tag{5.7}$$

For $0 \leq j \leq m$, the separable q-linearized polynomial $\mathcal{N}^{\circ j}$ has degree q^j, and its kernel is contained in $V_1 = \ker \mathcal{N}^{\circ m}$. Hence

$$\dim_{\mathbb{F}_q} \ker(\mathcal{N}^{\circ j}|_{V_1}) = j \qquad (0 \leq j \leq m).$$

Thus $\mathcal{N}|_{V_1}$ is nilpotent of index m on an m-dimensional space, so it is one nilpotent Jordan block. Since $\mathrm{Fr}_q = \mathcal{N} + \mathrm{id}$, arithmetic q-Frobenius acts as $J_m(1)$.

The splitting field of $\mathcal{A}_n(Q, 1)$ over $\mathbb{F}_q$ is finite, so its Galois group is cyclic and generated by arithmetic q-Frobenius. The computation above gives $\operatorname{disc}_Q \mathcal{A}_n(Q, v) = v^{Eq^m}$, a unit at $v = 1$. After choosing a prolongation of this place, Lemma 2.3 identifies its decomposition group with the Galois group of the specialized splitting field, with the same permutation action on the roots. The corresponding lift of arithmetic q-Frobenius is the required Frob. □

The fixed space $\ker(\mathrm{Frob} - I_m)$ is one-dimensional. In characteristic two, for every integer $j \geq 0$,

$$(I_m + J_m(0))^{2^j} = I_m + J_m(0)^{2^j}.$$

Consequently

$$\operatorname{ord}(\mathrm{Frob}) = 2^{\lceil \log_2 m \rceil}. \tag{5.8}$$

5.5. **The common root frame.** The integral expansions of Section 5.1 assemble into the following chain:

$$\Omega_n \xrightarrow{\text{integral expansions}} \{Q_j = z^{-D}W_j(z) : j \in \mathbb{Z}/N\} \xrightarrow{\text{mod } 2} V^\times \subset \mathbb{F}_{2^n}((z)) \xrightarrow{\mathrm{lc}_z} \mathbb{F}_{q^m}^\times. \tag{5.9}$$

Through it, inertia at infinity and Frobenius at $v = 1$, although obtained at different places, act on a single characteristic-zero root set.

Proposition 5.6 (compatible root identifications)**.** *For every odd $m \geq 1$, there is a single bijection*

$$\rho : \Omega_n \xrightarrow{\sim} V^\times, \qquad V \cong \mathbb{F}_q^m, \tag{5.10}$$

for which the following statements hold simultaneously:

(i) *the characteristic-zero inertia generator S at infinity acts $\mathbb{F}_q$-linearly with order L;*
(ii) *under the $\mathbb{F}_q$-linear identification $V \cong \mathbb{F}_{q^m}$ supplied by lc_z, the action of S is multiplication by an element of order L;*
(iii) *the lifted Frobenius* Frob *from $v = 1$ acts as $J_m(1)$;*
(iv) *$\mathcal{K} = \langle S, \mathrm{Frob}\rangle \leq G_n^{\mathrm{ar}}$, and under ρ every element of $\mathcal{K}$ acts on $V^\times$ by the restriction of an $\mathbb{F}_q$-linear transformation of V.*

Proof. Use the integral branches of Lemma 5.1 with $p = 2$ and residue field $\mathbb{F}_{2^n} = \mathbb{F}_{q^m}$. In random-cluster coordinates their roots are

$$Q_j = z^{-D}W_j(z), \qquad v = z^{-L}.$$

The change from $U_j = z^{-h}W_j$ multiplies roots and root differences by v, a unit for the Gauss valuation. Reduction therefore identifies the decomposition group at this chosen extension with the Galois group of $\widehat{\overline{Z}_n}$ over $\mathbb{F}_2(v)$, with the same permutation on the roots. By Proposition 5.3, the reduced roots together with zero form the space V. These first two arrows in (5.9) define the bijection ρ.

For a reduced root series $x \in V$, including $x = 0$, define

$$\mathrm{lc}_z(x) = z^D x\big|_{z=0}. \tag{5.11}$$

Thus the last arrow of (5.9) is evaluation at $\mathfrak{m} = (2, z)$ after multiplying by z^D. Addition and scalar multiplication of series make this map $\mathbb{F}_q$-linear. Its values on the nonzero roots are the distinct reductions of $-\zeta_N^j$, namely all of $\mathbb{F}_{q^m}^\times$. Hence $\mathrm{lc}_z : V \to \mathbb{F}_{q^m}$ is a linear isomorphism.

The inertia generator S of Section 3.4 is induced by $z \mapsto \xi z$ with $\xi \in \mu_L$ primitive; this action preserves the chosen Gauss place and fixes residue constants. Equation (5.1) gives

$$\mathrm{lc}_z(Sx) = \bar{\xi}^{-D}\,\mathrm{lc}_z(x) = \bar{\xi}^{-h}\,\mathrm{lc}_z(x). \tag{5.12}$$

Here $D = L + h$; odd-order roots of unity remain distinct modulo two, and $\gcd(h, L) = 1$. The multiplier has exact order L, proving (i)–(ii).

The residual splitting field contains $\mathbb{F}_q$: if $x \neq 0$ is a root, then every cx, with $c \in \mathbb{F}_q$, is a root, and their ratios recover the constants. Hence the subgroup over $\mathbb{F}_q(v)$ acts linearly on V. The reduction of S belongs to this subgroup by (5.12); the Frobenius element furnished by Proposition 5.5 belongs to it as well. Lemma 2.3 lifts both permutations through the same dyadic decomposition group to G_n^{ar}; we retain the name Frob for this lift. Under ρ, their characteristic-zero permutations are exactly their linear actions on $V^\times$, which proves (iii)–(iv). □

5.6. **Arithmetic of the inertia multiplier.** The multiplication action in Proposition 5.6 is the inertia action at infinity on Ω_n. The following lemma determines a scalar subgroup and a Sylow subgroup that constrain the induced action on an invariant block system of the roots.

Lemma 5.7 (scalar subgroup and Sylow normalizer)**.** *Let S be the inertia generator of Section 3.4, acting on $V \cong \mathbb{F}_{q^m}$ by multiplication as in Proposition 5.6(ii), and let p be the prime of Lemma 2.1(iii). Write $\Gamma L_1(q^m)$ for the semilinear group of maps $x \mapsto cx^{q^i}$ on $\mathbb{F}_{q^m}$, with $c \neq 0$ and $0 \leq i < m$. Then:*

(i) *S is irreducible on V, $\mathbb{F}_q[S] \cong \mathbb{F}_{q^m}$, and $\mathcal{Z} = \mathbb{F}_q^\times \leq \langle S \rangle$;*

(ii) *if Π is the p-primary subgroup of $\langle S \rangle$, then it is also the p-primary subgroup of $\mathbb{F}_{q^m}^\times$, $\Pi \in \mathrm{Syl}_p(\mathrm{GL}_m(q))$, and*

$$N_{\mathrm{GL}_m(q)}(\Pi) \leq \Gamma L_1(q^m). \tag{5.13}$$

In particular, this normalizer has odd order.

Proof. By Lemma 2.1(iv)–(v), $q - 1 \mid L$ and $\mathrm{ord}_L(q) = m$. The multiplier defining S has exact order L, so it has degree m over $\mathbb{F}_q$. This proves irreducibility and $\mathbb{F}_q[S] \cong \mathbb{F}_{q^m}$. Since $q - 1 \mid L$, the unique subgroup of order $q - 1$ in the multiplicative group $\mathbb{F}_{q^m}^\times$ lies in $\langle S \rangle$; it is $\mathcal{Z} = \mathbb{F}_q^\times$.

The equality $v_p(L) = v_p(N)$ identifies the p-primary subgroup of $\langle S \rangle$ with the unique p-primary subgroup of $\mathbb{F}_{q^m}^\times$. The order formula

$$|\mathrm{GL}_m(q)| = q^{m(m-1)/2} \prod_{i=1}^{m} (q^i - 1)$$

shows that it is Sylow in $\mathrm{GL}_m(q)$, because $\mathrm{ord}_p(q) = m$ excludes p from every factor $q^i - 1$ with $i < m$. Its characteristic subgroup of order p is generated by an element ϑ with $\mathbb{F}_q[\vartheta] = \mathbb{F}_{q^m}$. Every element normalizing Π therefore normalizes this field algebra. Conjugation induces an $\mathbb{F}_q$-automorphism of $\mathbb{F}_{q^m}$, hence a power of Frobenius. Composing with its inverse leaves an $\mathbb{F}_{q^m}$-linear map of a one-dimensional space, hence a scalar multiplication. This proves (5.13). Finally,

$$|\Gamma L_1(q^m)| = m(q^m - 1)$$

is odd. □

Part II. Permutation groups and full symmetry

6. Invariant block systems for the affine group

A block system (or system of imprimitivity) is a partition preserved by the group action. Throughout this section, $N = gL$ with g, L positive integers, $\mathcal{D}_{N,g}$ is the affine group (3.22) on $\mathbb{Z}/N$, and $T\colon j \mapsto j + g$ is the translation (3.23). This section classifies the affine-invariant block systems needed later and proves that every nonsingleton block has a nontrivial translation stabilizer under the arithmetic hypotheses of Lemma 6.4. For $g > 1$, the affine action is not transitive, so an invariant partition may have blocks of different sizes. Transitivity of T on the blocks in Theorem 6.2 forces equal sizes.

The following observation is the starting point of the orbitwise description of invariant block systems in the presence of semiregular abelian subgroups by Kovács–Malnič–Marušič–Miklavič [15, proof of Proposition 2.1].

Lemma 6.1 (blocks and orbits)**.** *Let a group I act on a set Ω, and let B be a block of an I-invariant partition, with setwise stabilizer I_B. Then $B \cap Ix = I_B x$ for every $x \in B$. In particular, if $I_B = 1$, then B meets each I-orbit in at most one point.*

Proof. If $y = \iota x \in B$ with $\iota \in I$, then ιB is a block containing $y \in B$, so $\iota B = B$ and $\iota \in I_B$. Conversely, $I_B x \subseteq B \cap Ix$. □

For the explicit affine action on $\mathbb{Z}/N$, the invariant block systems whose blocks are permuted transitively by T have the following sharper classification.

Theorem 6.2 (block systems for $\mathcal{D}_{N,g}$)**.** *Let $N = gL$ be odd. If a $\mathcal{D}_{N,g}$-invariant block system has k blocks and T acts transitively on them, then*

$$k \mid L, \qquad \gcd(k, g) = 1.$$

Moreover, its blocks are the congruence classes modulo k. Conversely, for every $k \mid L$ with $\gcd(k, g) = 1$, congruence modulo k is such a block system.

Proof. Transitivity makes the permutation induced by T a k-cycle, so $k \mid L$. Let B_0 be the block containing the label $0 \in \mathbb{Z}/N$, write

$$B_i = T^i B_0, \qquad i \in \mathbb{Z}/k,$$

and define $\phi : \mathbb{Z}/N \to \mathbb{Z}/k$ by $x \in B_{\phi(x)}$. Then

$$\phi(Tx) = \phi(x) + 1.$$

Every multiplier $M_u(j) = uj$ fixes B_0, and $M_u T M_u^{-1} = T^u$; hence

$$\phi(M_u x) = u\phi(x). \tag{6.1}$$

We choose a unit multiplier using the Chinese remainder theorem, which supplies t satisfying

$$\gcd(t, k) = 1, \qquad 1 + gt \in (\mathbb{Z}/N)^\times.$$

Indeed, at a prime $\ell \mid N$, require $t \not\equiv 0 \pmod{\ell}$ when $\ell \mid k$, and, when $\ell \nmid g$, also require $t \not\equiv -g^{-1} \pmod{\ell}$. There are at most two forbidden residues, so these conditions are compatible because ℓ is odd. Put $u = 1 + gt$. If $x \in B_i$, then

$$M_u(x) = T^{tx}(x).$$

Together with (6.1), this gives

$$ui \equiv i + tx \pmod{k},$$

and hence $t(x - gi) \equiv 0 \pmod{k}$. Thus $x \equiv gi \pmod{k}$. Taking $x = 1$ proves $\gcd(k, g) = 1$. The inclusions $B_i \subseteq \{x : x \equiv gi \pmod{k}\}$, together with equal cardinality, show that the blocks are exactly the congruence classes modulo k.

Conversely, if $k \mid L$ and $\gcd(k, g) = 1$, every map $j \mapsto uj + gb$ permutes the congruence classes modulo k, and T acts on them as addition by g, a k-cycle. □

Remark 6.3. The converse shows that the conclusion is sharp for the affine action alone. The condition $k \mid g$, obtained below from centered block sums for the hypercube, independently excludes all such block systems with $k > 1$. Without multiplication by the units of $\mathbb{Z}/N$, invariant block systems on whose blocks T acts transitively can also occur when k and g have a common divisor.

The preceding theorem concerns block systems on whose blocks translation acts transitively. We next give a condition that forces a nontrivial translation stabilizer in every nonsingleton block, without assuming transitivity on the blocks.

Lemma 6.4 (translation stabilizers)**.** *Suppose that g is odd and every prime dividing g divides L. Then every nonsingleton block of a $\mathcal{D}_{N,g}$-invariant block system on $\mathbb{Z}/N$ has nontrivial stabilizer in $\langle T \rangle$. Equivalently, each such block is a union of cosets of a nontrivial subgroup of $g\mathbb{Z}/N$.*

Proof. Put $I = \langle T \rangle$, and write I_B for the setwise stabilizer in I of a block B. By Lemma 6.1,

$$B \cap (I \cdot x) = I_B \cdot x \quad (x \in B).$$

Write $\mathcal{P}$ for the block system, B_x for the block containing x, and $\mathcal{P}/I$ for the set of I-orbits of blocks. The map

$$(\mathbb{Z}/N)/I \longrightarrow \mathcal{P}/I, \qquad I \cdot x \longmapsto I \cdot B_x$$

is well defined. Its fibers give the induced partition of $(\mathbb{Z}/N)/I \cong \mathbb{Z}/g$. Reduction $(\mathbb{Z}/N)^\times \to (\mathbb{Z}/g)^\times$ is surjective, so this partition is unit-invariant.

Suppose now that $I_B = 1$. Then B contains exactly one representative J_r of each residue class r it meets; write C for this set of residues, so $J_r \equiv r \pmod g$. Thus C is the fiber over $I \cdot B$ of the displayed quotient map. Let H_C and H_r be the setwise stabilizer of C and the stabilizer of $r \in C$ in $(\mathbb{Z}/g)^\times$, respectively. A unit fixing r sends C to an intersecting block and therefore preserves C; thus $H_r \le H_C$.

We first use oddness of g to show that pointwise fixation of C forces it to be a singleton. We then exclude a nontrivial action of H_C by a valuation argument using the hypothesis on the prime divisors of g.

Step 1. If H_C fixes every element of C, then C is a singleton. In this case $H_r = H_C$ for every $r \in C$. Put $M_r = g/\gcd(r, g)$. Then

$$H_r = \ker\big((\mathbb{Z}/g)^\times \longrightarrow (\mathbb{Z}/M_r)^\times\big)\,.$$

For odd g, these kernels distinguish the divisors M_r of g. Indeed, at a prime power $\ell^\nu \| g$, exponent zero gives the whole unit group; reduction modulo ℓ has a proper kernel because ℓ is odd; and the successive kernels for exponents $1, \dots, \nu$ decrease strictly. Hence M_r is constant on C. The unit group acts transitively on residues with a specified gcd with g. If $r' = wr$ were a second element of C, with $w \in (\mathbb{Z}/g)^\times$, then $wC \cap C \neq \varnothing$ would give $w \in H_C$, contrary to pointwise fixation.

Step 2. A nontrivial action of H_C is impossible. Otherwise, by taking a power of a suitable element, choose $\bar u \in H_C$ with a prime-length orbit $P > 1$ on some $r \in C$. Its order modulo M_r is P, so it has order P modulo some prime-power factor $\ell^d \| M_r$. Write

$$\nu = v_\ell(g), \qquad \mu = v_\ell(r), \qquad d = \nu - \mu > 0.$$

We choose a unit u modulo N, congruent to $\bar u$ modulo M_r, for which

$$v_\ell(u^P - 1) = d, \qquad \beta := v_\ell(u - 1) < d. \tag{6.2}$$

These valuations depend only on the prescribed residue modulo ℓ^{d+1}, because their target values are at most $d < d + 1$; throughout, they are computed on integer representatives with that residue.

If $P = \ell$, then $d \ge 2$, and the order condition gives $\beta = d - 1$. The lifting-the-exponent identity (2.4) gives (6.2). If $P \neq \ell$, then $\beta = 0$, since the kernel of reduction modulo ℓ is an ℓ-group. The derivative Pu^{P-1} is a unit modulo ℓ, so simple-root Hensel lifting [18, Chapter II, Proposition 7] gives exactly one lift of the prescribed residue modulo ℓ^{d+1} whose Pth power is one. Choose any other lift modulo ℓ^{d+1}. Since $\ell \mid g$, the hypothesis on L gives $v_\ell(N) > \nu \ge d$, so this digit is available. The Chinese remainder theorem preserves the residues modulo the other factors of M_r and chooses unit residues at the remaining primes dividing N.

The reduction of $u\bar u^{-1}$ modulo g fixes r, so it belongs to $H_r \le H_C$. Thus u still preserves C. It has the same P-cycle on r, since its action there depends only on its residue modulo M_r.

Because multiplication by u preserves C, it sends B to an I-translate. For one common $\tau_0 \in g\mathbb{Z}/N$, the representatives therefore satisfy

$$J_{ur'} = uJ_{r'} + \tau_0 \pmod N \quad (r' \in C).$$

Iterating around the P-cycle gives

$$(1 + u + \cdots + u^{P-1})\big((u - 1)J_r + \tau_0\big) \equiv 0 \pmod N. \tag{6.3}$$

Since $J_r \equiv r \pmod g$, its ℓ-valuation is $\mu < \nu$. The second factor has valuation $\beta + \mu < \nu$: the term τ_0 is divisible by g and therefore cannot cancel the lower-valuation term $(u - 1)J_r$. The first factor has valuation $d - \beta$, by (6.2). Their product has valuation

$$(d - \beta) + (\beta + \mu) = d + \mu = \nu < v_\ell(N),$$

contradicting (6.3). It follows that C, and hence B, is a singleton. □

Remark 6.5. Oddness of g cannot be omitted in this generality: for $N = 8$ and $g = 4$, the block system

$$\{\{0\},\{4\},\{2\},\{6\},\{1,3\},\{5,7\}\}$$

is $\mathcal{D}_{8,4}$-invariant, but $\{1,3\}$ has trivial stabilizer under translation by 4.

Remark 6.6. The hypothesis that every prime dividing g also divides L is likewise necessary. Take $N = 15$, $g = 3$, $L = 5$, and multiplication by 11. Its orbit partition is

$$\{\{0\},\{3\},\{6\},\{9\},\{12\},\{1,11\},\{2,7\},\{4,14\},\{5,10\},\{8,13\}\}.$$

This block system is $\mathcal{D}_{15,3}$-invariant, while every displayed two-element block has trivial stabilizer under translation by 3.

7. Primitivity and full symmetry

We now rule out every nontrivial block system for the hypercube arithmetic monodromy group. The proof separates according to $g = \gcd(N, E)$. Centered block sums give $k \mid g$; when $g > 1$, the affine translation-stabilizer lemma verifies the required stabilizer condition. For a G_n^{ar}-invariant block system, let b denote its common block size and let $k = N/b$ be its number of blocks. If $g = 1$, the first divisibility proves primitivity immediately. If $g > 1$, the residual linear action makes inertia transitive on the blocks, and the affine classification gives $\gcd(k, g) = 1$. These are the nested regimes of the introduction; when N is prime, even the centered sums are unnecessary (Remark 7.8). Once primitivity is established, the geometric transposition from Section 4 yields full symmetry.

7.1. **Centered block sums.** The coefficient of U^{N-1} in $R_n(U, v)$ is E, so $-E/N$ is the mean of its roots. To constrain a block system, we cancel the polar terms in its block sums. The following theorem formulates the classical no-poles constancy principle [22, Lemma 33.9.3, Tag 0BUG] for block traces, including its integral divisibility consequence.

Theorem 7.1 (bounded block traces and divisibility)**.** *Let*

$$F(X, Y) = X^N + \alpha(Y)X^{N-1} + \cdots \in \mathbb{Z}[Y][X]$$

be monic and separable, with transitive arithmetic Galois group Γ*. Let* $\mathcal{P}$ *be a* Γ*-invariant partition into* k *blocks of size* b*, so* $N = bk$*, and put*

$$\Theta_B = N \sum_{\omega \in B} x_\omega + b\alpha(Y). \tag{7.1}$$

Suppose $w(\Theta_B) \geq 0$ *for every* $B \in \mathcal{P}$*, at one place* w *of the splitting field above* $Y = \infty$*. Then every* Θ_B *is an algebraic constant. If either geometric monodromy is transitive on* $\mathcal{P}$*, or one* Θ_B *has rational residue at* w*, then*

$$\Theta_B = 0, \qquad \sum_{\omega \in B} x_\omega = -\frac{\alpha(Y)}{k} \in \mathbb{Z}[Y] \quad (B \in \mathcal{P}). \tag{7.2}$$

In particular, k *divides every coefficient of* α*. If* $[\alpha] \in (\mathbb{Z}/N\mathbb{Z})[Y]$ *has additive order* L*, then* $L \mid b$*.*

Proof. The arithmetic group permutes the blocks, so

$$\mathcal{H}(t, Y) = \prod_{B \in \mathcal{P}} (t - \Theta_B) \in \mathbb{Q}(Y)[t]. \tag{7.3}$$

Monicity of F makes its roots, the Θ_B, and the coefficients of $\mathcal{H}$ integral over $\mathbb{Z}[Y]$. Thus $\mathcal{H} \in \mathbb{Z}[Y][t]$. All its coefficients are bounded at w; a polynomial in Y bounded above infinity is constant. Hence $\mathcal{H} \in \mathbb{Z}[t]$, and each Θ_B is an algebraic constant.

Geometric monodromy fixes these constants. If it is transitive on the blocks, they are all equal. Alternatively, w is trivial on the algebraic constant field, so reduction is injective there. A rational

residue therefore makes one Θ_B rational, and arithmetic transitivity makes every Θ_B equal to it. In either case their common value is zero, since

$$\sum_{B\in\mathcal{P}} \Theta_B = N\sum_{\omega} x_\omega + kb\alpha(Y) = 0. \tag{7.4}$$

This proves the trace identity. Each root sum is integral over $\mathbb{Z}[Y]$, so $-\alpha(Y)/k \in \mathbb{Z}[Y]$. Finally $b[\alpha] = 0$ gives $L \mid b$. □

For an expansion

$$Nx_\omega + \alpha(Y) = \lambda(\omega)\Psi + \varepsilon_\omega, \qquad w(\varepsilon_\omega) \geq 0, \tag{7.5}$$

phase cancellation $\sum_{\omega\in B}\lambda(\omega) = 0$ makes the centered block sum bounded. A sufficient condition is a setwise stabilizer σ of B with $\lambda(\sigma\omega) = c\lambda(\omega)$, $c \neq 1$: then

$$\sum_{\omega\in B}\lambda(\omega) = \sum_{\omega\in B}\lambda(\sigma\omega) = c\sum_{\omega\in B}\lambda(\omega), \qquad \text{hence } \sum_{\omega\in B}\lambda(\omega) = 0. \tag{7.6}$$

If all remainders have positive valuation, every centered block sum has residue zero, so the rational-residue alternative applies. For absolutely irreducible F, geometric transitivity applies directly.

Corollary 7.2 (block traces under the defect bound)**.** *Let $\mathcal{M}$ satisfy the hypotheses of Proposition 3.7. Consider an arithmetic block system with k blocks of size b. If every block has zero phase sum $\sum_{\eta\in B}\eta = 0$, then*

$$\sum_{\eta\in B} U_\eta = -\frac{e}{k}, \qquad L \mid b, \qquad k \mid g. \tag{7.7}$$

The phase sums vanish if every block has a nontrivial setwise stabilizer in infinity inertia; this holds whenever $b > g$. In particular, if $g = 1$, arithmetic monodromy is primitive.

Proof. By Corollary 3.8, cancellation of the phases cancels the whole common polar series in each block sum. Thus $r\sum_{\eta\in B} U_\eta + be$ is bounded. Absolute irreducibility supplies geometric transitivity, and Theorem 7.1 gives the trace identity and $L \mid b$, since $[e] \in \mathbb{Z}/r\mathbb{Z}$ has order L. As $r = bk = gL$, also $k \mid g$.

Inertia acts on the phases by the faithful character $\xi \mapsto \xi^{-h}$ of μ_L, so a nontrivial stabilizer gives (7.6). By Lemma 6.1, a block with trivial stabilizer meets each of the g inertia orbits in at most one point; hence $b > g$ suffices. If $g = 1$, every proper nonsingleton block would have $b > g$ and $r = L \mid b$, a contradiction. □

For polynomial covers $f(X) - Y = 0$, Dujella, Gusić and Tichy [8, Corollary 1] give a related coefficient obstruction: a monic $f(X) = X^N + aX^{N-1} + \cdots \in \mathbb{Z}[X]$ with $\gcd(N, a) = 1$ is indecomposable. Goodall, Jouve and Sereni apply it to uniform matroids [10, proof of Theorem 4.7(2)]. Here Corollary 7.2 obtains $k \mid g$ from the defect bound and phase cancellation.

Section 8 combines the primitivity statement with the transposition of Proposition 4.1 to prove the general criterion.

7.2. **The induced action on blocks.** For the hypercube with $g > 1$, we next prove that inertia is transitive on the blocks. The following proposition isolates the fixed-line argument.

Proposition 7.3 (transitivity on a block system from fixed points)**.** *Let a finite group $\mathcal{K}$ preserve a block system on a finite set Ω, with k blocks of odd size. Suppose $\Pi \in \mathrm{Syl}_p(\mathcal{K})$ acts trivially on the block set and $N_{\mathcal{K}}(\Pi)$ has odd order. If $\mathrm{Frob} \in \mathcal{K}$ has 2-power order and some $\mathcal{Z} \leq \mathcal{K}$ preserves and acts transitively on $\mathrm{Fix}_\Omega(\mathrm{Frob})$, then $\mathcal{Z}$ is transitive on the blocks. In particular, the trivial-action hypothesis on Π holds whenever $p > k$.*

Proof. Let J be the kernel of the action on the blocks. Then $\Pi \le J \triangleleft \mathcal{K}$, and Π is a Sylow p-subgroup of J. The Frattini argument gives

$$\mathcal{K} = JN_{\mathcal{K}}(\Pi).$$

Indeed, every $\mathcal{K}$-conjugate of Π is a Sylow subgroup of J and can be conjugated back to Π by an element of J. The induced group $\mathcal{K}/J$ is therefore a quotient of $N_{\mathcal{K}}(\Pi)$ and has odd order. The image of the 2-element Frob is trivial, so Frob fixes every block setwise.

Each block has odd size, whereas every nontrivial $\langle \mathrm{Frob} \rangle$-orbit has even size. Hence every block contains a Frob-fixed point. Sending a point of $\mathrm{Fix}_\Omega(\mathrm{Frob})$ to its block is therefore a surjective $\mathcal{Z}$-equivariant map. Transitivity of its source proves transitivity of the block set. Finally, if $p > k$, then $p \nmid k!$, so the permutation action on the k blocks kills Π. □

Corollary 7.4 (transitivity of inertia on a block system)**.** *Use the prime p of Lemma 2.1(iii) and the common root representation of Proposition 5.6. If a G_n^{ar}-invariant block system has $k < p$ blocks of odd size, then the inertia subgroup $\langle S \rangle$ acts transitively on its blocks.*

Proof. Put $\mathcal{K} = \langle S, \mathrm{Frob} \rangle$. Let Π be the p-primary subgroup of $\langle S \rangle$. Lemma 5.7 makes Π a Sylow p-subgroup of $\mathrm{GL}(V)$, and its normalizer there has odd order. Hence $\Pi \in \mathrm{Syl}_p(\mathcal{K})$ and $N_{\mathcal{K}}(\Pi)$ has odd order.

For the regular unipotent $\mathrm{Frob} \sim J_m(1)$, (5.8) gives Frob 2-power order, while $\Lambda = \ker(\mathrm{Frob} - I_m)$ is one-dimensional. Thus $\Lambda^\times = \mathrm{Fix}_{V\setminus\{0\}}(\mathrm{Frob})$ is a regular orbit of $\mathcal{Z} = \mathbb{F}_q^\times$, which centralizes Frob. Lemma 5.7 and Proposition 5.6 give $\mathcal{Z} \le \langle S \rangle \le \mathcal{K}$. Proposition 7.3 therefore makes $\mathcal{Z}$, and hence $\langle S \rangle$, transitive on the blocks. □

7.3. **Arithmetic primitivity.**

Proposition 7.5 (arithmetic primitivity)**.** *For every $n \ge 2$, the action of G_n^{ar} on Ω_n is primitive.*

Proof. Put $\Gamma = G_n^{\mathrm{ar}}$, and suppose that a Γ-invariant block system has common block size $b > 1$ and $k = N/b$ blocks. We prove $k = 1$.

If $g = 1$, then $b > g$, so Corollary 7.2 gives $k = 1$ at once. Suppose now that $g > 1$. Lemma 2.1(i)–(ii) verifies the arithmetic hypothesis of Lemma 6.4. By Proposition 3.9, the block system is invariant under $\mathcal{D}_{N,g}$, and inertia $I = \langle S \rangle$ acts on the labels as $\langle T \rangle$ (3.23). Lemma 6.4 makes the stabilizer I_B nontrivial for every block B. Corollary 7.2 therefore gives

$$L \mid b, \qquad k \mid g. \tag{7.8}$$

Let $p > g$ be the prime of Lemma 2.1(iii). Then $k \le g < p$, and b is odd because N is odd. Corollary 7.4 makes $\langle S \rangle = \langle T \rangle$ transitive on the blocks. Theorem 6.2 now gives $\gcd(k, g) = 1$. Together with $k \mid g$, this forces $k = 1$. □

Remark 7.6 (the three block-system tests when $n = 6$). Using Table 2.2, the parameters are

$$(N, E, \sigma, g, L, h, D) = (63, 192, 129, 3, 21, 43, 64), \qquad (a, m, q, \kappa, p) = (2, 3, 4, 64, 7).$$

Thus $V \cong \mathbb{F}_4^3 \cong \mathbb{F}_{64}$, inertia is multiplication by an element of order 21, and $\mathrm{Frob} \sim J_3(1)$ has order four. The following are $\mathcal{D}_{63,3}$-invariant block systems on the affine label set, used as models for the three tests.

(i) The residue classes modulo 7 give seven blocks of size 9. They are affine-invariant and T is transitive on them, so the affine classification permits them. Centered sums exclude any global realization because $21 \nmid 9$.

(ii) A three-block system of block size 21 passes the centered divisibility test. The residue classes modulo 3 form an affine-invariant example, but $T(j) = j + 3$ fixes its blocks. For any G_6^{ar}-invariant three-block system, the residual argument would make T transitive; the affine classification would then require $\gcd(3, 3) = 1$, which is impossible.

(iii) The cosets of $21\mathbb{Z}/63\mathbb{Z}$ and of $9\mathbb{Z}/63\mathbb{Z}$ have block sizes 3 and 7, respectively. Both systems are affine-invariant, and centered sums exclude them because neither size is divisible by $L = 21$.

7.4. **From primitivity to full symmetry.** We finish with the standard orbital-graph form of the primitive-plus-transposition argument; see [7, §3.2 and Theorem 3.3A].

Lemma 7.7 (transpositions and orbital graphs)**.** *Let $H \triangleleft \Gamma \leq \mathrm{Sym}(\Omega)$, and let $t = (x\ y) \in H$. Define*

$$\mathcal{G}_\Gamma(t) = \big(\Omega, \{\{\gamma x, \gamma y\} : \gamma \in \Gamma\}\big). \tag{7.9}$$

The orbit of the ordered pair is closed under reversal because t interchanges x and y. If $\mathrm{Comp}(\mathcal{G}_\Gamma(t))$ is the set of connected components, then

$$\langle t^\Gamma \rangle = \prod_{B \in \mathrm{Comp}(\mathcal{G}_\Gamma(t))} \mathrm{Sym}(B). \tag{7.10}$$

In particular, if $\mathcal{G}_\Gamma(t)$ is connected, then $H = \Gamma = \mathrm{Sym}(\Omega)$.

Proof. The conjugates t^Γ are exactly the edge transpositions of $\mathcal{G}_\Gamma(t)$. Edge transpositions generate the full symmetric group on each connected component, proving (7.10). They all lie in H by normality. If the graph is connected, then $\mathrm{Sym}(\Omega) \leq H \leq \Gamma \leq \mathrm{Sym}(\Omega)$. □

Proof of Theorem 1.1. Put $H = G_n^{\mathrm{geo}}$ and $\Gamma = G_n^{\mathrm{ar}}$. Equation (2.5) gives $H \triangleleft \Gamma$, and Proposition 4.1 gives a transposition $t_p \in H$. The components of $\mathcal{G}_\Gamma(t_p)$ form a Γ-invariant block system. Since Γ is primitive by Proposition 7.5, this system consists either of singletons or of one block. The edge of t_p excludes singletons. Hence the graph is connected, and Lemma 7.7 gives $H = \Gamma = S_N = S_{2^n-1}$. Remark 2.2 gives the same group over $K(Y)$ for every field K of characteristic zero. □

Remark 7.8 (the prime-degree case). When $N = 2^n - 1$ is prime, Proposition 7.5 is not needed: a transitive permutation group of prime degree is primitive, so Theorem 1.1 already follows from the transitivity of Section 2.1, the transposition of Proposition 4.1 with $p = N$, and Lemma 7.7. Likewise, when $g = 1$, the proof of Theorem 1.3 uses Lemma 6.1 but no other result from Sections 5–6.

Part III. Criteria and further families

8. Density criteria and graph applications

We first prove Theorem 1.3 and its consequences, then develop congruence traces for odd prime-power rank and finite-field certificates for the graph applications.

Proof of Theorem 1.3. By Section 2.1, $T_{\mathcal{M}}$ is absolutely irreducible, so both root actions are transitive. Here $g = \gcd(r, e) = 1$, so Corollary 7.2 makes the arithmetic action primitive. Choose any prime $p \mid r$; because $g = 1$, Proposition 4.1 supplies a transposition in geometric monodromy. Normality and Lemma 7.7 now show that the geometric and arithmetic groups are both S_r, and Remark 2.2 gives the same group over $K(Y)$ for every field K of characteristic zero. □

Proof of Corollary 1.4. Let d be the valency of G. Since G is connected and noncomplete, one has $d \geq 2$. A finite connected vertex-transitive graph of valency at least two is 2-connected; see Watkins [23]. Thus the cycle matroid of G is connected, of rank $r = M - 1$, size $e = E$, and positive nullity. Corollary 3.6 shows that singletons are the unique minimum-defect nonempty nonspanning edge sets, and the coprimality hypothesis is $\gcd(r, e) = \gcd(M-1, E) = 1$. Theorem 1.3 applies. □

The examples of Corollary 1.4 in Appendix A satisfy its valency form of the coprimality condition: cycles have valency two; $K_{t,t}$ has $M = 2t$ and $d = t$, and $K_{t,t,t}$ has $M = 3t$ and $d = 2t$, so the condition reduces to $\gcd(2t-1, t) = 1$ and $\gcd(3t-1, t) = 1$; the cube, dodecahedron,

Heawood, Pappus, Desargues and Tutte–Coxeter graphs are cubic of order $M \not\equiv 1 \pmod 3$; and the octahedron and icosahedron have even order with valency 4 and 5 coprime to $M-1=5$ and 11. Petersen and its complement have $\gcd(M-1,E)=3$ and are treated by orbit traces below. Clebsch, Coxeter and Hoffman–Singleton are treated by finite-field certificates in Section 8.3.

Corollary 8.1 (bipartite vertex-transitive graphs)**.** *Let G be a connected simple bipartite vertex-transitive graph on M vertices, of power-of-two valency $d \geq 2$. Then its arithmetic and geometric Tutte monodromy groups are both S_{M-1} over every field of characteristic zero.*

Proof. The graph is 2-connected by Watkins [23], so its cycle matroid is connected. Since G is regular and bipartite, its two parts have the same size, say t. Thus $M=2t$ and $E=dt$, and

$$\gcd(M-1,E)=\gcd(2t-1,dt)=1,$$

because d is a power of two and $\gcd(2t-1,t)=1$. Lemma 3.4 supplies the singleton-defect hypothesis, so Theorem 1.3 applies. □

For example, the Cartesian products $C_4 \square C_{2t}$, $t \geq 3$, have monodromy S_{8t-1}. They are bipartite, vertex-transitive, and 4-regular. An edge in the C_4 direction belongs to three 4-cycles, while an edge in the other direction belongs to two, so these graphs are not edge-transitive.

Corollary 8.2 (projective geometries)**.** *Let q_0 be a prime power and $r \geq 2$. Put $e=(q_0^r-1)/(q_0-1)$. If $\gcd(r,e)=1$, then the arithmetic and geometric Tutte monodromy of $\mathrm{PG}(r-1,q_0)$ is S_r over every field of characteristic zero. In particular, this applies to the Fano matroid.*

Proof. Projective geometries are connected. Put $e_j=(q_0^j-1)/(q_0-1)$. A set of rank j contains at most e_j points, hence its defect is at least $f(j)=ej-re_j$. The sequence $f(j)$ is strictly concave, because $f(j+1)-f(j)=e-rq_0^j$ strictly decreases. Now $f(1)=e-r=s$, and, for $r \geq 3$,

$$f(r-1)-s=[q_0(r-2)-r]e_{r-1}+2r-2>0.$$

For $r=3$ the right side is $(q_0-1)^2$; for $r \geq 4$ its bracketed coefficient is nonnegative. Concavity therefore gives $f(j)>s$ for $2 \leq j < r$. Rank-one sets are singletons, which have defect s; these are also the only nonempty nonspanning sets when $r=2$. Theorem 1.3 applies. For the Fano matroid, $(r,e)=(3,7)$. □

For $q_0=2$, the coprimality condition is $\gcd(r,2^r-1)=1$. This is exactly $g=1$ for the cube in dimension $n=r$, so binary projective geometries satisfy this criterion in the same dimensions where the cube's primitivity proof needs no characteristic-two argument.

8.1. **Prime-power ranks and congruence traces.** For odd prime-power rank, the cyclotomic multipliers determine all possible block systems. The following elementary observation supplies the required classification.

Lemma 8.3 (multiplier-invariant partitions)**.** *Let p be an odd prime and $a \geq 1$. Every partition of $\mathbb{Z}/p^a\mathbb{Z}$ into sets of equal size, preserved by all unit multipliers, is the congruence partition modulo p^c for some $0 \leq c \leq a$.*

Proof. Write the common size as $d=p^{a-c}$. The cases $c=0,a$ are immediate; assume $0<c<a$. The block B_0 containing zero is fixed by every multiplier. The orbit of a label of p-adic valuation $t<c$ has size $(p-1)p^{a-t-1} \geq (p-1)d > d$. Hence $B_0=p^c\mathbb{Z}/p^a\mathbb{Z}$, by cardinality.

Let $B \neq B_0$ and let Λ be its stabilizer in $(\mathbb{Z}/p^a\mathbb{Z})^\times$. Write Ω_t for the labels of valuation t, and $\mathcal{J}=\{t : B \cap \Omega_t \neq \varnothing\}$; thus $t<c$. The stabilizer of any point of $B \cap \Omega_t$ has order p^t and lies in Λ. Moreover $B \cap \Omega_t$ is a single Λ-orbit: any multiplier carrying one of its points to another must preserve B. With $t_*=\max \mathcal{J}$, orbit–stabilizer gives

$$d=|\Lambda|\sum_{t\in\mathcal{J}} p^{-t}=\frac{|\Lambda|}{p^{t_*}}\sum_{t\in\mathcal{J}} p^{t_*-t}.$$

The first factor is an integer and the second is prime to p. Since d is a power of p, the second factor equals one. Thus $\mathcal{J} = \{t\}$ and $|\Lambda| = p^{a-c+t}$. The unit group modulo p^a is cyclic; its unique subgroup of this order is the kernel of reduction modulo p^{c-t}. Its orbit through $j \in \Omega_t$ is $j + p^c\mathbb{Z}/p^a\mathbb{Z}$, proving the claim. □

Theorem 8.4 (prime-power trace criterion)**.** *Let $\mathcal{M}$ be a connected matroid of rank $r = p^a$, where p is an odd prime and $a \geq 2$, and size $e > r$. Put*

$$s = e - r, \qquad g = \gcd(r, e) = p^b, \qquad L = r/g, \qquad h = s/g, \qquad 0 \leq b \leq a.$$

Suppose that

$$\delta_{\mathcal{M}}(A) \geq s \qquad (\varnothing \neq A \subseteq \mathcal{E},\ r(A) < r). \tag{8.1}$$

Equality is allowed for nonsingletons. Label the roots at infinity by $U_j = z^{-h}W_j(z)$, where $v = z^{-L}$ and $W_j(0) = -\zeta_r^j$, as in Proposition 3.9. Every nontrivial arithmetic block system is the congruence partition modulo p^c for some $1 \leq c \leq \min(b, a-1)$, and its block sums satisfy

$$\sum_{j \equiv i \pmod{p^c}} U_j = -\frac{e}{p^c} \qquad (i \in \mathbb{Z}/p^c\mathbb{Z}). \tag{8.2}$$

Consequently, arithmetic monodromy is primitive if, for every such c, at least one candidate trace differs from $-e/p^c$. In particular, nonconstancy of the trace with $i = 0$ for each c suffices.

Proof. The defect bound gives simplicity and strict uniform density by Proposition 3.7. Absolute irreducibility gives transitivity. The decomposition group at infinity contains every unit multiplier on the phase labels (Proposition 3.9), so Lemma 8.3 makes every nontrivial block system a congruence partition modulo p^c, with $1 \leq c < a$. Each block has size $p^{a-c} > 1$ and zero phase sum. Corollary 7.2 therefore gives its exact trace $-e/p^c$ and $p^c \mid g$, proving $c \leq b$. □

For $b = 0$, there are no candidate nontrivial systems. For $b = 1$, the only candidate is congruence modulo p, whose classes are exactly the infinity-inertia orbits. The argument also covers $b = a$, when $L = 1$: phase cancellation follows from the multiplier classification.

Trace coefficient certificates. Sums of Puiseux branches over blocks and the selection of their coefficients by roots of unity occur in the polynomial moment problem; see Gavrilov and Pakovich [9, §3.3, equations (29)–(35)]. Here we test the prescribed block-trace identity (8.2). For such c, put $k = p^c$, $d = r/k$, and $\tau_c(v) = \sum_{j \equiv 0 \pmod{k}} U_j$. Compute in the finite algebra

$$\mathcal{A}_c = \mathbb{Z}[1/r][t]/(t^d + 1).$$

Here $t^r = -1$ and $\Phi_W(t, 0) = -r/t$ is a unit, so there is a unique $\mathcal{W}_c(t, z) \in \mathcal{A}_c[\![z]\!]$ with $\mathcal{W}_c(t, 0) = t$ and $\Phi(\mathcal{W}_c, z) = 0$. Evaluation at the roots of $t^d + 1$ gives precisely the branches in τ_c. The algebra trace of a representative of degree less than d is d times its constant coefficient. Thus

$$\tau_c(v) = dz^{-h}[t^0]\mathcal{W}_c(t, z) = d\sum_{q \geq 0}[t^0 z^{h+qL}]\mathcal{W}_c(t, z)\, v^{-q}. \tag{8.3}$$

Under the singleton-uniqueness hypothesis, the $q = 0$ term is $-e/k$. The coefficient $[t^0]$ is taken after reduction modulo $t^d + 1$. The label set is invariant under all multipliers and translation by g; Proposition 3.9 therefore places its trace in $\mathbb{Q}((v^{-1}))$, explaining the exponent filter in (8.3). Writing $\mathcal{W}_c = \sum_{n \geq 0} a_n z^n$, the recursion is

$$a_0 = t, \qquad a_n = \frac{t}{r}[z^n]\Phi\left(\sum_{j<n} a_j z^j, z\right) \pmod{t^d + 1} \quad (n \geq 1). \tag{8.4}$$

For any prime $\ell \neq p$, the same derivative is a unit over $\mathbb{F}_\ell[t]/(t^d+1)$, so unique lifting commutes with reduction. A nonzero reduced coefficient in (8.3) with $q \geq 1$ certifies nonconstancy in characteristic

zero. A constant term different from $-e/k$, or unequal constant terms of two coset traces, also excludes the partition by (8.2). Only the terms of Φ through z^{h+qL} are needed. For $c < b$, this trace sums p^{b-c} inertia orbits.

For a single inertia orbit, let $W(z)$ be the branch with $W(0) = -1$. Its normalized roots are $\xi^h W(\xi^{-1}z)$, where $\xi^L = 1$; the corresponding U-trace is

$$\tau_0(v) = L \sum_{j \equiv h \pmod{L}} [z^j]W(z)\, z^{j-h}, \qquad v = z^{-L}. \tag{8.5}$$

This gives the tests used in the following examples. A nonzero reduced coefficient with $j > h$ and $j \equiv h \pmod{L}$, at a prime not dividing r, certifies a nonconstant trace.

8.2. **Petersen and its complement.**

Corollary 8.5. *For the Petersen graph P and its complement $J(5,2)$, the arithmetic and geometric Tutte monodromy groups are both S_9 over every field of characteristic zero.*

Proof. Both graphs are connected, noncomplete, and arc-transitive. Their parameters are

G	r	e	g	L	h
P	9	15	3	3	2
$J(5,2)$	9	30	3	3	7.

Corollary 3.6 supplies the defect hypothesis of Theorem 8.4.

For Petersen, the defining edge-subset sum gives the weighted prefix

$$\begin{aligned}\Phi_P(W,z) = {} & W^9 + 1 + 15z^2W^8 + 15z^3 \\ & + (105W^7 + 10W)z^4 + (12W^5 + 15W^2)z^5 + O(z^6).\end{aligned} \tag{8.6}$$

For $\eta^9 = -1$, substitution gives

$$\begin{aligned}W_\eta(z) = {} & \eta - \frac{5}{3}z^2 + \frac{5}{3}\eta z^3 + \frac{5\eta^8 + 10\eta^2}{9}z^4 \\ & + \frac{4\eta^6 + 5\eta^3}{3}z^5 + O(z^6).\end{aligned} \tag{8.7}$$

Summing on the orbit $\eta \in \{-1, -\omega, -\omega^2\}$, where $\omega^3 = 1$ and $\omega \neq 1$, and dividing by z^2 gives

$$\tau_P(v) = -5 - v^{-1} + O(v^{-2}). \tag{8.8}$$

The remainder uses inertia invariance, which permits only powers of $z^3 = v^{-1}$ in the trace.

For the complement, the corresponding prefix is

$$\begin{aligned}\Phi_{J(5,2)}(W,z) = {} & W^9 + (1+z^3)^{30} + 30z^7W^8 + 10z^8W \\ & + (240W + 30W^7)z^{11} + 5W^6z^{12} + 30W^2z^{13} + O(z^{14}).\end{aligned} \tag{8.9}$$

Appendix B gives the finite coefficient calculations for both prefixes. Start with the unperturbed roots $\eta(1+z^3)^{10/3}$. All nonscalar perturbations have degree at least seven, so first-order implicit differentiation determines their effects through degree thirteen. The terms $30z^7W^8$ and $30z^{13}W^2$ contribute respectively $-10z^7/3$ and $10\eta^3 z^{13}/3$ to W_η. The other terms are scalar multiples of η, η^2, η^8, or η^7, with coefficients in $\mathbb{Q}[\![z^3]\!]$, and sum to zero on the three-root orbit. Since $\sum \eta^3 = -3$, division by z^7 yields

$$\tau_{J(5,2)}(v) = -10 - 10v^{-2} + O(v^{-3}). \tag{8.10}$$

Both traces are nonconstant, so arithmetic monodromy is primitive.

For the geometric transpositions, the exact residue certificates in Appendix B have the form $T_G(X, Y_0) = (X - X_0)^2 C(X)$ in $\mathbb{F}_\ell[X]$, with C squarefree and $C(X_0) \neq 0$. Their derivative values are

G	ℓ	X_0	Y_0	$C(X_0)$	$T_{XX}(X_0, Y_0)$	$T_Y(X_0, Y_0)$
P	11	9	2	8	5	10
$J(5,2)$	17	9	13	14	11	10.

(8.11)

The Jacobian of $(T_G, T_{G,X})$ is invertible at each residual point. The simultaneous Hensel and quadratic-factor argument in the proof of Proposition 4.1 therefore gives an algebraic characteristic-zero branch value with exactly one double root and all other roots simple. Its geometric inertia is a transposition. Primitivity and Lemma 7.7 give both groups S_9, and Remark 2.2 gives the stated base change. □

8.3. **Finite-field certificates.** The next applications obtain primitivity from the order of a specialization Frobenius. We use the following elementary observation.

Lemma 8.6 (prime-order obstruction)**.** *Let $\Gamma \leq S_r$ be transitive and contain an element of prime order ℓ. A Γ-invariant partition into k blocks of size d, with $k, d > 1$, is impossible if $\ell > \max(k, d)$.*

Proof. Such an element fixes every block, since $\ell > k$, and every point in each block, since $\ell > d$. It would therefore be the identity. □

Corollary 8.7 (Clebsch)**.** *The arithmetic and geometric Tutte monodromy groups of the Clebsch graph are both S_{15} over every field of characteristic zero.*

Proof. The graph is connected, noncomplete and arc-transitive, with $(r, e, g) = (15, 40, 5)$. Corollary 3.6 supplies the singleton-defect hypothesis, and Proposition 4.1, with $p = 3$, gives a geometric transposition. The squarefree specialization $R_G(U, 1)$ modulo five has irreducible factor degrees $1, 7, 7$ (Appendix B.2). Lemma 2.3 and the specialization principle following it give an arithmetic element of order seven. The only possible nontrivial block systems have block size and number $(3, 5)$ or $(5, 3)$; Lemma 8.6 excludes both. Primitivity, normality, and Lemma 7.7 give S_{15}. Remark 2.2 gives the base change. □

For the next two graphs, we combine a prime cycle with a geometric discriminant certificate. We record the standard normalization argument in a form that also permits a single residue value to certify the sign.

Lemma 8.8 (normalized discriminant)**.** *Let $D \in \mathbb{Z}[v]$ be nonzero of even degree and leading coefficient a. If D is a square in $\overline{\mathbb{Q}}(v)$, then $D/a = H^2$ for a monic $H \in \mathbb{Q}[v]$. For every prime $\ell \nmid a$, one has $H \in \mathbb{Z}_{(\ell)}[v]$. Consequently, either a nonsquare polynomial $\overline{D/a} \in \mathbb{F}_\ell[v]$ or a nonsquare value $\overline{D/a}(v_0) \in \mathbb{F}_\ell$ excludes geometric squareness.*

Proof. A rational-function square root of a polynomial has no finite poles, so it is polynomial. The monic square root of D/a is unique and Galois-invariant, hence rational. It is integral over the integrally closed ring $\mathbb{Z}_{(\ell)}[v]$, hence belongs to that ring. Reduction and evaluation preserve squares. □

Corollary 8.9 (Coxeter and Hoffman–Singleton)**.** *The arithmetic and geometric Tutte monodromy groups of the Coxeter and Hoffman–Singleton graphs are respectively S_{27} and S_{49} over every field of characteristic zero.*

Proof. Both graphs are connected, noncomplete and arc-transitive, with $(r, e) = (27, 42)$ and $(49, 175)$. Their Tutte polynomials are absolutely irreducible. Appendix B gives the following squarefree specializations of $R_G(U, v)$:

Graph	Prime	v_0	Factor degrees	Frobenius power
Coxeter	7	-1	$1^6, 8, 13$	8
Hoffman–Singleton	5	2	$1, 13, 35$	35

By Lemma 2.3 and the specialization principle following it, these factor patterns give elements of arithmetic monodromy Γ. The indicated powers are 13-cycles. A nontrivial block system in degree 27 has block size and number $(3, 9)$ or $(9, 3)$; in degree 49 it has $(7, 7)$. Lemma 8.6 therefore makes Γ primitive. Jordan's prime-cycle theorem [7, Theorem 3.3E] gives $A_r \le \Gamma$ in both cases.

Put $D(v) = \operatorname{disc}_U R_G(U, v)$. For Coxeter, (3.10) gives degree 390 and leading coefficient $a = -27^{27} \equiv 1 \pmod 7$. The squarefree specialization in (B.9) has discriminant $D(-1) \equiv 6 \pmod 7$, a nonsquare. For Hoffman–Singleton, the same initial-form calculation gives

$$\deg D = 6048, \qquad a = 49^{49}, \qquad D(1) \equiv 2 \pmod 3. \tag{8.12}$$

Here $a \equiv 1 \pmod 3$, so $\overline{D/a}(1) = 2$ is a nonsquare. Lemma 8.8 supplies an odd geometric permutation in each case.

Let H be geometric monodromy. It is transitive and normal in Γ. The intersection $H \cap A_r$ is normal in the simple group A_r. If it were trivial, H would inject into Γ/A_r, of order at most two, contradicting transitivity. Thus $A_r \le H$. The odd geometric element gives $H = \Gamma = S_r$, and Remark 2.2 gives the stated base change. □

Remark 8.10 (a Coxeter trace certificate). Coxeter also illustrates the modular coefficient test for Theorem 8.4. Here $(r, e, g, L, h) = (27, 42, 3, 9, 5)$. In the weighted reduction modulo seven, set $t = z^7$. The branch with $W(0) = -1$ satisfies $[t^2]W = [t^{11}]W = 0$ and $[t^{20}]W = 5$, as verified by (B.8). These are the first three candidate exponents in the filter $7d \equiv 5 \pmod 9$. The characteristic-zero coefficient $[z^{140}]W$ lies in $\mathbb{Z}_{(7)}$ and reduces to five. Hence

$$[v^{-15}]\tau_0(v) = [z^{135}]\tau_0(z^{-9}) \equiv 9 \cdot 5 = 3 \pmod 7, \tag{8.13}$$

giving an independent proof of arithmetic primitivity.

8.4. **Scope of the method.**

(i) The parity calculation extends to a Cayley graph of $\mathbb{F}_2^d$ with any generating direction set $\Sigma \subseteq \mathbb{F}_2^d \setminus \{0\}$. For $\mathcal{T} \subseteq \Sigma$, put $\operatorname{rk}(\mathcal{T}) = \dim_{\mathbb{F}_2} \operatorname{span}(\mathcal{T})$. Translation-fixed edge sets give

$$Q\,\overline{\widehat{Z}_G}(Q, v) = \sum_{\mathcal{T} \subseteq \Sigma} Q^{2^{d-\operatorname{rk}(\mathcal{T})}} v^{|\mathcal{T}|2^{d-1}},$$

a 2-linearized polynomial. The Lucas-theorem refinement to a single $q = 2^a$ and the regular-unipotent calculation at $v = 1$ use the particular direction set of the hypercube and do not follow from this formula alone.

(ii) For arc-transitive graphs with $g > 1$, the first root-dependent correction still centers the block sums. If a prime divisor of the rank does not divide g, Proposition 4.1 also gives a geometric transposition. For $r = p^a$ with p odd and $a \ge 2$, Theorem 8.4 requires only the bound $\delta_{\mathcal{M}}(A) \ge s$ and proves primitivity when the candidate congruence partitions are excluded by their trace identities, for $1 \le c \le \min(v_p(g), a - 1)$. For ranks with several prime factors, the unit-multiplier partitions require further analysis; the Clebsch example instead uses the prime-order obstruction of Lemma 8.6. In even rank, the initial roots are $\eta^r = -1$, and the labels $-\zeta_r^j$ of Proposition 3.9 are replaced accordingly.

9. Uniform matroids

For uniform matroids, two binomial identities control all finite branch values. The proof uses the Morse-polynomial method applied to cycles by Goodall–Jouve–Sereni [10, §4.2], together with a disjointness statement for the two sets of critical values. This proves Theorem 1.2 for every $2 \le r < e$, extending the sufficiently-large-size conclusion of [10, Theorem 4.7(1)].

Put $s = e - r \ge 1$ and $C = \binom{e}{r} = \binom{e}{s}$. Write

$$A_r(U) = \sum_{j=0}^{r-1} \binom{e}{j} U^{r-j}, \qquad A_s(v) = \sum_{j=0}^{s-1} \binom{e}{j} v^{s-j}, \qquad B(v) = C + A_s(v). \tag{9.1}$$

The rank function of $U_{r,e}$ gives the additive separation

$$R(U,v) := T_{U_{r,e}}(U+1, v+1) = A_r(U) + B(v). \tag{9.2}$$

The binomial recurrence yields

$$U(U+1)A_r'(U) = (rU - s)A_r(U) + rCU, \tag{9.3}$$

$$v(v+1)B'(v) = (sv - r)B(v) + rC. \tag{9.4}$$

For (9.3), comparison of successive coefficients uses $(j+1)\binom{e}{j+1} = (e-j)\binom{e}{j}$; the lowest coefficient uses $(s+1)\binom{e}{r-1} = rC$. Interchanging r and s and adding the constant C gives (9.4). The complementary sums in (9.1) also give

$$A_r(U) + B(U^{-1}) = \frac{(U+1)^e}{U^s}. \tag{9.5}$$

Lemma 9.1 (critical points and values)**.** *The polynomial A_r has $r-1$ simple critical points, and their critical values are pairwise distinct. If α is a critical point, then $\alpha \notin \{0, -1, s/r\}$ and*

$$A_r(\alpha) = -\frac{rC\alpha}{r\alpha - s}, \qquad \alpha(\alpha+1)A_r''(\alpha) = -\frac{rsC}{r\alpha - s}. \tag{9.6}$$

No critical value of B is the negative of a critical value of A_r.

Proof. First, $A_r'(0) = \binom{e}{r-1} \ne 0$. Evaluating (9.3) and its derivative at -1 gives

$$A_r(-1) = -\frac{rC}{e}, \qquad A_r'(-1) = \frac{rsC}{e(e-1)} = \binom{e-2}{r-1} \ne 0.$$

At $U = s/r$, (9.3) precludes $A_r' = 0$. For a critical point α, the same identity gives the first formula in (9.6). Differentiating it and using $A_r'(\alpha) = 0$ gives

$$\alpha(\alpha+1)A_r''(\alpha) = r\big(A_r(\alpha) + C\big) = -\frac{rsC}{r\alpha - s} \ne 0.$$

Thus all critical points are simple. Their values are distinct because $\alpha \mapsto -rC\alpha/(r\alpha - s)$ is an injective fractional-linear map.

When $s = 1$, B' is a nonzero constant. When $s \ge 2$, the same argument with r and s interchanged applies to A_s. A critical point β of B satisfies $s\beta - r \ne 0$, and (9.4) gives

$$B(\beta) = -\frac{rC}{s\beta - r}.$$

If $A_r(\alpha) + B(\beta) = 0$ for critical points α, β, then

$$\frac{\alpha}{r\alpha - s} + \frac{1}{s\beta - r} = 0, \qquad \text{so} \qquad \alpha\beta = 1.$$

Equation (9.5) would then give $(\alpha+1)^e = 0$, contrary to $\alpha \ne -1$. □

Proposition 9.2 (uniform discriminant)**.** *For $2 \le r < e$, the polynomial $\Delta_{r,e}(v) = \operatorname{disc}_U R(U,v)$ is squarefree, of degree $s(r-1)$ and leading coefficient $(-1)^{r(r-1)/2}r^r$. Every finite branch value of the projection to the v-line has exactly one ramification point, of ramification index two.*

Proof. Let $\alpha_1, \dots, \alpha_{r-1}$ be the critical points of A_r. Since A_r is monic, the resultant formula gives

$$\Delta_{r,e}(v) = (-1)^{r(r-1)/2} r^r \prod_{i=1}^{r-1} \big(B(v) + A_r(\alpha_i)\big). \tag{9.7}$$

The distinct critical values in Lemma 9.1 imply that different factors have disjoint zero sets. The disjointness of the critical values of B and $-A_r$ implies that each factor is squarefree. Each factor is monic of degree s, proving the discriminant assertions.

At a zero v_0 of $\Delta_{r,e}$, exactly one critical point α_i satisfies $R(\alpha_i, v_0) = 0$. At that point,

$$R_U = 0, \qquad R_{UU} = A_r''(\alpha_i) \neq 0, \qquad R_v = B'(v_0) \neq 0.$$

Thus α_i is the unique double root of $R(U, v_0)$, the other roots are simple, and the projection has ramification index two there. Monicity in U makes the affine projection finite, so these are all its finite branch values. □

Proof of Theorem 1.2. The matroid $U_{r,e}$ is connected for $0 < r < e$, so its Tutte polynomial is absolutely irreducible by Section 2.1. Its geometric monodromy is therefore transitive. Over $\mathbb{C}$, loops around the finite branch values generate this group: the loop around infinity is the inverse of their product. Proposition 9.2 makes each of these generators a transposition. Their transposition graph is connected by transitivity, and its edge transpositions generate S_r, as in the proof of Lemma 7.7. The arithmetic group contains the geometric group and lies in S_r, so it too is S_r. Remark 2.2 gives the conclusion over every characteristic-zero constant field. □

For rank one, the polynomial is linear in X, giving the trivial group S_1. Thus the same conclusion holds for every connected uniform matroid of positive rank.

10. A separability criterion for Brylawski polynomials

The cube proof uses the first root-dependent correction, which is stronger information than the binomial weighted initial form alone. Combining that initial form with the Brylawski identity gives a separability criterion for a wider class of polynomials. We now prove Theorem 1.5. When $\ell \nmid r$, separability already follows from the leading discriminant coefficient $(-1)^{r(r-1)/2}r^r$ in (3.10); the theorem addresses primes dividing the rank.

A nonzero polynomial $F(X,Y)$ is called an (e,r)-Brylawski polynomial if, for some $c \neq 0$,

$$(Y-1)^r F\left(\frac{Y}{Y-1}, Y\right) = cY^e. \tag{10.1}$$

It is *normalized* when $c = 1$. Every rank-r Tutte polynomial on e elements is normalized. For a short modern derivation of the Brylawski coefficient relations from the Tutte hyperbola, see Beke, Csáji, Csikvári and Pituk [1]. Goodall, Jouve and Sereni prove over an arbitrary unique factorization domain that factors of a Brylawski polynomial are again Brylawski polynomials, with additive parameters and sharp degree bounds [10, Lemma 2.5 and Proposition 2.2]. In the monic normalized case needed here, the identity for factors and its first-order consequence have a direct proof.

Lemma 10.1 (monic factors and expansion at $Y = 1$)**.** *Let $F \in K[Y][X]$ be monic of bidegree (r,s) over a field K, and suppose that F satisfies the identity (1.3). If $F = AB$, with both factors monic in X, and A has bidegree (a,b), then*

$$(Y-1)^a A\left(\frac{Y}{Y-1}, Y\right) = Y^{a+b}. \tag{10.2}$$

If $a \geq 1$ and $A = X^a + a_{a-1}(Y)X^{a-1} + \cdots$, then

$$a_{a-1}(1) = b \quad \text{in } K. \tag{10.3}$$

In characteristic $\ell > 0$, the condition $A_X = 0$ therefore forces $\ell \mid a$ and $\ell \mid b$.

Proof. Denote the left side of (10.2) by $A^{\#}(Y)$, and define $B^{\#}$ using the X-degree $r - a$. These are polynomials with

$$A^{\#}B^{\#} = Y^{r+s}, \qquad A^{\#}(1) = B^{\#}(1) = 1,$$

where the values at 1 follow from monicity. Unique factorization therefore gives $A^{\#} = Y^d$ and $B^{\#} = Y^{r+s-d}$. Since $\deg_Y B = s - b$, the substitution also gives

$$\deg A^{\#} \leq a + b, \qquad \deg B^{\#} \leq (r - a) + (s - b).$$

The two upper bounds sum to the degree of the product, so both are equalities. Thus $d = a + b$, proving (10.2).

If $a = 0$, monicity gives $A = 1$ and $b = 0$, so the remaining assertions are immediate. For $a \geq 1$, set $Y = 1 + t$ and retain the first-order terms:

$$(1 + t)^a + t(1 + t)^{a-1}a_{a-1}(1 + t) + O(t^2) = (1 + t)^{a+b}.$$

Comparison of the coefficients of t gives (10.3). Finally, if $A_X = 0$ in characteristic ℓ, monicity gives $\ell \mid a$, and the coefficient a_{a-1} is zero. Hence $b = 0$ in K as well. □

Proof of Theorem 1.5. Work first over $K = \mathbb{F}_\ell$. By hypothesis,

$$\mathrm{in}_{(s,r)} F = X^r + Y^s.$$

This initial form is squarefree: its Y-derivative is $sY^{s-1} \neq 0$ and is coprime to it. If F had a repeated irreducible factor over $K(Y)$, Gauss's lemma for monic factors would put the complete factorization in $K[Y][X]$, and multiplicativity of weighted initial forms would make the square of a nonconstant polynomial divide $X^r + Y^s$. Thus the irreducible factors of F are distinct.

It remains possible in positive characteristic that one of those distinct factors is inseparable in X. Suppose that $\gcd(F, F_X) \neq 1$, choose a monic irreducible $A \mid \gcd(F, F_X)$, and write $F = AB$ in $K[Y][X]$. Since $A \nmid B$, reduction of

$$F_X = A_X B + AB_X$$

modulo A gives $A \mid A_X$. Degree then forces

$$A_X = 0. \tag{10.4}$$

Put $a = \deg_X A$ and $b = \deg_Y A$. Weighted degree for $w(X) = s$ and $w(Y) = r$, together with monicity and degree additivity, gives

$$\begin{aligned} w(A) + w(B) &= rs, \\ w(A) \geq sa, &\quad w(B) \geq s(r - a), \\ w(A) \geq rb, &\quad w(B) \geq r(s - b). \end{aligned}$$

Both pairs of lower bounds already sum to rs, so all four inequalities are equalities and $sa = rb$. Write $r = gL$ and $s = gh$ with $\gcd(L, h) = 1$. Then

$$(a, b) = (Lu, hu), \qquad 1 \leq u \leq g. \tag{10.5}$$

Lemma 10.1 applies to this factorization. The inseparability condition (10.4) forces its subleading coefficient to vanish, whereas the first-order expansion of the Brylawski identity gives $a_{a-1}(1) = b$. Thus $\ell \mid b = hu$. Yet (1.4) gives $\ell \nmid h$ and $1 \leq u \leq g < \ell$, a contradiction. Hence $\gcd(F, F_X) = 1$, proving separability.

Return to characteristic zero and put $\Delta_F(Y) = \operatorname{disc}_X F(X,Y)$. For a nonzero rational polynomial P, write

$$v_\ell^{\mathrm{G}}(P) = \min_i v_\ell([Y^i]P)$$

for its Gauss ℓ-valuation. Equation (3.10) gives the degree and leading coefficient of Δ_F from the binomial initial form. Separability modulo ℓ says that the Gauss ℓ-valuation of Δ_F is zero. Define the monic normalization

$$\Delta_F^{\mathrm{mon}}(Y) = \frac{(-1)^{r(r-1)/2}}{r^r}\Delta_F(Y). \tag{10.6}$$

If Δ_F^{mon} were a square in $\mathbb{Q}(Y)$, unique factorization would write it as $R(Y)^2$ with $R \in \mathbb{Q}[Y]$. Comparison of Gauss valuations in $\Delta_F = (-1)^{r(r-1)/2} r^r R^2$ would give

$$0 = r\, v_\ell(r) + 2v_\ell^{\mathrm{G}}(R).$$

The first term is odd and the second even, a contradiction.

If the monic polynomial Δ_F^{mon} were a square in $\overline{\mathbb{Q}}(Y)$, unique factorization would give a unique monic square root in $\overline{\mathbb{Q}}[Y]$. Every automorphism of $\overline{\mathbb{Q}}/\mathbb{Q}$ would fix that root, placing it in $\mathbb{Q}[Y]$, contrary to the preceding paragraph. Finally, every nonzero rational constant is a square in $\overline{\mathbb{Q}}$. Thus Δ_F and Δ_F^{mon} have the same square class over $\overline{\mathbb{Q}}(Y)$, proving (1.5). □

Corollary 10.2 (strictly uniformly dense matroids)**.** *Let $\mathcal{M}$ be a connected strictly uniformly dense matroid of rank $r > 0$ and nullity $s > 0$. If an odd prime ℓ satisfies (1.4), then $T_{\mathcal{M}} \bmod \ell$ is separable in X. If r and $v_\ell(r)$ are also odd, then*

$$\operatorname{disc}_X T_{\mathcal{M}}(X,Y) \notin \overline{\mathbb{Q}}(Y)^{\times 2}.$$

Equivalently, its geometric root action contains an odd permutation.

Proof. Every Tutte polynomial is a normalized Brylawski polynomial, and Proposition 3.1 supplies its binomial weighted initial form. In particular, the coefficient of its highest X-power X^r is one, so $T_{\mathcal{M}}$ is monic in X. □

For the cube, the prime p of Lemma 2.1(iii) satisfies the separability hypotheses. Whenever $v_p(N)$ is odd, the corollary also gives an odd geometric permutation independently of Theorem 1.1; for example, this applies to $n = 3$ and $p = 7$.

Appendix A. Coverage of graph and matroid families

Tables A.1 and A.2 collect the applications of the results in this paper. Every listed object has arithmetic and geometric Tutte monodromy S_r over every field of characteristic zero. For a connected graph, $r = |V| - 1$ and $e = |E|$; throughout the tables, $g = \gcd(r, e)$.

Graph or family	Parameters and condition	Mechanism and reference	
Hypercubes Q_n, $n \geq 2$	$r = 2^n - 1$, $e = n2^{n-1}$; all g	Two-place comparison	Thm. 1.1[†]
Cycles C_M, $M \geq 3$	$r = M - 1$, $e = M$	Critical values	Thm. 1.2[*†]
Connected noncomplete simple arc-transitive graphs	$r = \|V\| - 1$, $e = \|E\|$; $g = 1$	Coprime density	Cor. 1.4[†]
Connected simple bipartite vertex-transitive graphs	$d = 2^a \geq 2$, $M = 2t$; $r = 2t - 1$, $e = 2^a t$	Coprime density	Cor. 8.1[†]
$K_{t,t}$, $t \geq 2$	$r = 2t - 1$, $e = t^2$; $g = 1$	Coprime density	Cor. 1.4[†]
$K_{t,t,t}$, $t \geq 2$	$r = 3t - 1$, $e = 3t^2$; $g = 1$	Coprime density	Cor. 1.4[†]
Cube, octahedron	$(r, e) = (7, 12), (5, 12)$	Coprime density	Cor. 1.4[†]
Dodecahedron, icosahedron	$(r, e) = (19, 30), (11, 30)$	Coprime density	Cor. 1.4
Heawood, Pappus	$(r, e) = (13, 21), (17, 27)$	Coprime density	Cor. 1.4
Desargues, Tutte–Coxeter	$(r, e) = (19, 30), (29, 45)$	Coprime density	Cor. 1.4
Petersen	$(r, e, g) = (9, 15, 3)$	Orbit traces	Cor. 8.5[†]
$J(5, 2)$, complement of Petersen	$(r, e, g) = (9, 30, 3)$	Orbit traces	Cor. 8.5[†]
Clebsch	$(r, e, g) = (15, 40, 5)$	Prime order	Cor. 8.7
Coxeter	$(r, e, g) = (27, 42, 3)$	Prime cycle; sign	Cor. 8.9
Hoffman–Singleton	$(r, e, g) = (49, 175, 7)$	Prime cycle; sign	Cor. 8.9

Table A.1. Graph coverage. The mechanism column identifies the route used here. Cycles have cycle matroid $U_{M-1,M}$. Finite certificates for the five final graphs are in Appendix B.

[*] The full cycle family was proved by Goodall–Jouve–Sereni [10, Theorem 4.5].

[†] Bohn–Cameron–Müller computed the arithmetic group over $\mathbb{Q}(Y)$ for all biconnected graphs on at most ten vertices [2, preceding Conjecture 9]. For the marked families, their range is $n \leq 3$, $M \leq 10$, $t \leq 5$ for $K_{t,t}$, $t \leq 3$ for $K_{t,t,t}$, and $|V| \leq 10$ for the arc-transitive and bipartite vertex-transitive rows.

Matroid or family	Parameters and condition	Mechanism and reference	
Uniform $U_{r,e}$	$2 \leq r < e$; all g	Critical values	Thm. 1.2[‡]
Projective geometry $\mathrm{PG}(r-1, q_0)$	$e = (q_0^r - 1)/(q_0 - 1)$; $r \geq 2$, $g = 1$	Coprime density	Cor. 8.2
Fano matroid	$(r, e) = (3, 7)$	Coprime density	Cor. 8.2
Connected matroid with the singleton-defect hypothesis	$e > r \geq 2$; $g = 1$	Coprime density	Thm. 1.3

Table A.2. Matroid coverage. The singleton-defect hypothesis is the inequality $\delta_{\mathcal{M}}(A) \geq e - r$ for every nonempty nonspanning set, with equality exactly for singletons.

[‡] For each fixed $r \geq 2$, Goodall–Jouve–Sereni proved arithmetic monodromy S_r over $\mathbb{Q}(Y)$ for $U_{r,e}$ with e sufficiently large [10, Theorem 4.7(1)].

Appendix B. Exact certificates for the graph extensions

For the modular calculations, let a finite ℓ-group P act by graph automorphisms. It acts on every set of edge subsets of specified rank and cardinality; each nonfixed orbit has size divisible by ℓ. Hence

$$R_G(U,v) \equiv \sum_{\substack{A\subseteq E(G)\\ PA=A}} U^{r-r(A)}v^{|A|-r(A)} \pmod{\ell}. \tag{B.1}$$

The fixed subsets are exactly the unions of the edge orbits of P.

B.1. **Petersen and the Johnson graph $J(5,2)$.** Realize both graphs on the two-element subsets of $\{0,1,2,3,4\}$: adjacency means disjointness for Petersen and intersection for its complement. One exact way to reconstruct either polynomial is to sum over the 115975 set partitions π of the ten vertices:

$$Z_G(Q,v) = \sum_{\pi}(Q)_{|\pi|}(1+v)^{e(\pi)}, \qquad R_G(U,v) = \frac{Z_G(Uv,v)}{Uv^{10}}. \tag{B.2}$$

Here $(Q)_j = Q(Q-1)\cdots(Q-j+1)$, and $e(\pi)$ is the number of graph edges whose endpoints lie in one block of π. The first identity follows by grouping Potts colorings by their equal-color partition; the second follows by substituting $Q = Uv$ in the edge-subset sum. In particular, differentiation before specializing v verifies the T_Y entries of (8.11).

For Petersen, the defining sum over 2^{15} edge subsets also gives

$$\begin{aligned} T_P(X,Y) = {} & X^9 + 6X^8 + 21X^7 + 56X^6 + (12Y+114)X^5 \\ & + (70Y+170)X^4 + (30Y^2+170Y+180)X^3 \\ & + (15Y^3+105Y^2+240Y+120)X^2 \\ & + (10Y^4+65Y^3+171Y^2+168Y+36)X \\ & + Y^6+9Y^5+35Y^4+75Y^3+84Y^2+36Y. \end{aligned} \tag{B.3}$$

The terms in (8.6) count, in order, the empty and full edge sets, single edges, deletion of one edge, two-edge sets, isolation of one vertex, the twelve five-cycles, and isolation of an adjacent vertex pair.

For $J(5,2)$, direct enumeration of the nonempty vertex subsets gives

Number of vertices	1	2	3	4	5	6	7	8	9	10
Maximum induced edges	0	1	3	6	8	12	15	19	24	30
Number attaining maximum	10	30	30	5	60	5	30	30	10	1

An edge subset of rank ρ and size a contributes weighted degree $10\rho-3a$. Applying the table to its connected components shows that weighted degree at most thirteen permits exactly the following configurations:

Term in Φ	Edge subsets counted
W^9	The empty set.
$(1+z^3)^{30}$, through z^{13}	Delete at most four edges; the graph stays connected.
$30z^7W^8$	One edge.
$10z^8W$	Isolate one vertex and retain all other edges.
$30z^{11}W^7$	A triangle: twenty triples incident with one vertex of the underlying K_5, and its ten triangles.
$240z^{11}W$	Isolate one vertex and delete one of the remaining twenty-four edges.
$5z^{12}W^6$	One of the five K_4's formed by the full stars of K_5.
$30z^{13}W^2$	Isolate two adjacent vertices and retain all other edges.

This proves the complete prefix (8.9); (B.2) supplies an independent exact check.

The branch certificates are

$$\begin{aligned} T_P(X,2) &= (X+2)^2 C_P(X) && \text{in } \mathbb{F}_{11}[X], \\ C_P(X) &= X^7+2X^6+9X^5+X^4+10X^3+2X^2+9X+7, \\ T_{J(5,2)}(X,13) &= (X-9)^2 C_J(X) && \text{in } \mathbb{F}_{17}[X], \\ C_J(X) &= X^7+5X^6+5X^5+9X^4+9X^3+15X^2+3X+12. \end{aligned} \tag{B.4}$$

Euclidean division gives $\gcd(C_P, C_P') = \gcd(C_J, C_J') = 1$; evaluation and differentiation give the entries of (8.11). These identities can all be checked from (B.2), or from (B.3) for P.

B.2. **Clebsch modulo five.** Realize Clebsch as the Cayley graph of $\mathbb{F}_2^4$ with directions e_1, e_2, e_3, e_4, and $e_1+e_2+e_3+e_4$. The linear map taking these five directions cyclically to one another is a graph automorphism of order five, with eight edge orbits of size five. Its 256 fixed edge subsets, counted by rank deficiency, give

$15-r(A)$	0	1	2	5	6	7	10	11	15
Number	96	103	9	21	17	1	6	2	1

By (B.1), the specialization $f(U) = R_G(U,1) \bmod 5$ is therefore

$$\begin{aligned} f(U) &= U^{15}+2U^{11}+U^{10}+U^7+2U^6+U^5+4U^2+3U+1 \\ &= (U+1)A(U)B(U), \\ A(U) &= U^7+2U^6+3U^5+3U^4+3U^3+U^2+4U+2, \\ B(U) &= U^7+2U^6+4U^5+2U^4+3U^3+2U^2+3. \end{aligned} \tag{B.5}$$

Multiplication verifies the factorization, and $\gcd(f,f')=1$. For each $H \in \{A,B\}$, the identities $U^{5^7} \equiv U \pmod{H}$ and $\gcd(H, U^5-U)=1$ prove irreducibility. Thus the specialization Frobenius has type (1)(7)(7) and order seven.

B.3. **Coxeter modulo seven.** A concrete model of the Coxeter graph has as vertices the 28 three-element subsets of $\mathbb{Z}/7\mathbb{Z}$ other than the seven triples $\{i, i+1, i+3\}$; two vertices are adjacent when disjoint. Translation by one has six edge orbits of size seven. For this cyclic action, all nonfixed orbits of edge subsets have cardinality seven, so

$$R_G(U,v) \equiv \sum_{\substack{A \subseteq E(G) \\ A+1=A}} U^{27-r(A)} v^{|A|-r(A)} \pmod 7. \tag{B.6}$$

The fixed subsets are the 64 unions of those six edge orbits. Computing their ranks as $28-c(A)$ gives the complete reduction

$$\begin{aligned} R_G(U,v) \equiv\ & U^{27}+3U^{21}v+3U^{20}+3U^{15}v^2+2U^{14}v \\ & +3U^7v^8+v^{15}+3U^{13}+U^9v^3+2U^8v^2 \\ & +3Uv^9+2U^7v+3v^8+U^6+3U^2v^3-Uv^2+3v \pmod 7. \end{aligned} \tag{B.7}$$

Thus the graph definition, its rotation, and these 64 rank counts suffice to recover every modular input to the Coxeter proof.

Substitute $U = z^{-5}W$, $v = z^{-9}$ and multiply by z^{135}. The resulting polynomial uses only $t = z^7$. Its simple root with $W(0) = -1$ satisfies

$$\begin{aligned} W = 6 &+ 4t^3+4t^4+3t^5+2t^6+2t^8+6t^9+2t^{10} \\ &+ 5t^{12}+3t^{13}+2t^{14}+5t^{15}+5t^{16}+5t^{17}+6t^{18}+5t^{20}+O(t^{21}) \quad \text{in } \mathbb{F}_7[\![t]\!]. \end{aligned} \tag{B.8}$$

Substitution into (B.7) makes the weighted polynomial zero modulo t^{21}, which verifies the trace coefficient used in (8.13).

Substituting $U = X - 1$ and $v = -1$ gives

$$\begin{aligned} T_G(X,0) &= \left(\prod_{i=0}^{5}(X+i)\right) c_8(X)c_{13}(X), \\ c_8(X) &= X^8 + 6X^7 + 4X^6 + 6X^4 + 2X^2 + 3X + 5, \\ c_{13}(X) &= X^{13} + X^{12} + 4X^{11} + 6X^9 + 4X^7 + X^6 \\ &\quad + 3X^5 + 3X^4 + 5X^3 + 4X^2 + 6X + 3 \qquad \text{in } \mathbb{F}_7[X]. \end{aligned} \tag{B.9}$$

The factors c_8, c_{13} are irreducible. This is verified by the usual finite-field test: for a monic polynomial f of degree d, check $X^{7^d} \equiv X \pmod f$ and $\gcd(f, X^{7^{d/\ell}} - X) = 1$ for every prime $\ell \mid d$. The specialization identity follows by multiplication using (B.7). Its discriminant is 6 in $\mathbb{F}_7$, giving the Coxeter sign certificate. These checks use only the polynomials displayed here.

B.4. **Hoffman–Singleton modulo five and three.** Use the following model of the Hoffman–Singleton graph. Its vertices are $p_{i,j}$ and $q_{i,j}$ for $i, j \in \mathbb{F}_5$, with edges

$$p_{i,j}p_{i,j+1}, \qquad q_{i,j}q_{i,j+2}, \qquad p_{i,j}q_{k,j+ik} \quad (i,j,k \in \mathbb{F}_5).$$

The automorphisms

$$\begin{aligned} \alpha &: p_{i,j} \mapsto p_{i+1,j}, \qquad q_{k,l} \mapsto q_{k,l+k}, \\ \beta &: p_{i,j} \mapsto p_{i,j-i}, \qquad q_{k,l} \mapsto q_{k+1,l} \end{aligned}$$

generate a group of order 125, whose three edge orbits have sizes $25, 25, 125$. The ranks and nullities of their eight unions give

$$R_G(U,v) \equiv U^{49} + 2U^{29}v^5 + U^9v^{10} + v^{126} + 2v^{101} + v^{76} \pmod 5. \tag{B.10}$$

At $v = 2$ this becomes

$$R_G(U,2) = U^{49} + 4U^{29} + 4U^9 + 4 = (U+1)c_{13}(U)c_{35}(U) \quad \text{in } \mathbb{F}_5[U]. \tag{B.11}$$

In ascending coefficient order the two factors are

$$\begin{aligned} c_{13} &: (4,2,3,4,0,4,3,3,4,2,1,4,1,1), \\ c_{35} &: (1,1,1,3,4,4,0,2,3,1,3,4,4,3,4,4,4,3, \\ &\qquad 4,2,1,4,1,0,1,0,2,0,3,0,4,3,2,4,3,1). \end{aligned}$$

Their product verifies (B.11). The usual finite-field tests, with exponents 5^{13} and 5^{35} and the proper prime-divisor subfield tests, prove irreducibility; the specialization has derivative gcd one.

For the reduction modulo three, number $p_{i,j}$ by $5i+j$ and $q_{i,j}$ by $25+5i+j$, using representatives $0, \dots, 4$. The following permutations preserve the edge set and generate a group of order nine:

$$\begin{aligned} \rho = {} & (0\ 27\ 5)(1\ 29\ 6)(2\ 9\ 45)(3\ 8\ 18)(4\ 7\ 31) \\ & (10\ 15\ 20)(11\ 16\ 21)(12\ 37\ 30)(14\ 44\ 46)(17\ 43\ 35) \\ & (19\ 38\ 41)(22\ 49\ 40)(24\ 32\ 36)(33\ 34\ 39)(42\ 48\ 47), \\ \sigma_3 = {} & (0\ 5\ 27)(1\ 49\ 12)(2\ 4\ 43)(6\ 22\ 30)(7\ 35\ 9) \\ & (10\ 15\ 20)(11\ 36\ 46)(13\ 26\ 23)(14\ 16\ 24)(17\ 45\ 31) \\ & (19\ 42\ 39)(21\ 32\ 44)(29\ 40\ 37)(33\ 38\ 48)(34\ 41\ 47). \end{aligned}$$

Their edge orbits have sizes 1, four copies of 3, and eighteen copies of 9. Summing over their 2^{23} unions in (B.1) and setting $v = 1$ gives $f \in \mathbb{F}_3[U]$ of degree 49, with ascending coefficient vector

$$\begin{aligned} f : (&1, 0, 2, 2, 2, 2, 0, 2, 0, 0, 2, 0, 1, 2, 1, 1, 2, 0, 1, 2, 0, 2, 0, 1, 0, \\ &0, 2, 0, 0, 2, 1, 2, 2, 2, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1). \end{aligned} \tag{B.12}$$

Euclidean division gives $\gcd(f, f') = 1$, and the resultant gives

$$\operatorname{disc}(f) = 2 \in \mathbb{F}_3. \tag{B.13}$$

As an independent sign check, the irreducible factor degrees are $1, 2, 3, 10, 15, 18$, giving Frobenius sign $(-1)^{49-6} = -1$. Together with the leading coefficient in (8.12), this proves the geometric sign used in Corollary 8.9.

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